

1. Title Page

Project FORESIGHT

AI-Powered Demand & Inventory Intelligence Platform

Notebook 01: Data Engineering & Validation

Project Information

Item	Details
Author	Mohit Kumar
Internship	Zidio Development
Project Type	End-to-End Data Analytics & Machine Learning Project
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2. Business Problem

Retail businesses operate in highly dynamic environments where customer demand fluctuates due to seasonality, promotions, holidays, pricing strategies, and changing consumer behavior. Inaccurate demand estimation often leads to two major business challenges:

- **Overstocking**, which increases inventory holding costs and ties up working capital.
- **Stockouts**, which result in lost sales, dissatisfied customers, and reduced customer satisfaction.

Traditional inventory planning methods often rely on historical averages or manual decision-making, making it difficult for retailers to respond quickly to changing demand patterns.

Project FORESIGHT aims to address these challenges by developing an AI-powered demand and inventory intelligence platform. The project combines data engineering, analytics, machine learning, and business intelligence to generate demand forecasts, monitor inventory health, identify stock-related risks, and support data-driven inventory decisions.

Building reliable analytical and forecasting models requires high-quality data. Therefore, the first objective of this project is to transform the raw M5 Forecasting datasets into clean, validated, and business-ready datasets, which is the primary purpose of this notebook.

Key Takeaway

Effective demand forecasting and inventory management are fundamental to retail success. Project FORESIGHT addresses these challenges by building a data-driven platform that transforms raw retail data into actionable business intelligence.

3. Notebook Objective

The primary objective of this notebook is to build a robust data engineering pipeline that transforms the raw M5 Forecasting datasets into clean, validated, and business-ready datasets for downstream analytics and machine learning.

Specifically, this notebook will:

- Load the raw M5 datasets.
- Understand the structure and quality of each dataset.
- Perform data engineering and feature creation.
- Build standardized business-ready datasets.
- Validate the engineered datasets for consistency and accuracy.

Export the processed datasets for use in subsequent notebooks. The datasets generated in this notebook will serve as the foundation for exploratory data analysis, feature engineering, demand forecasting, inventory intelligence, Power BI dashboard development, and Streamlit application deployment throughout Project FORESIGHT.

Key Takeaway

The primary goal of Notebook 01 is to transform the raw M5 datasets into clean, validated, and business-ready datasets that will support all subsequent stages of Project FORESIGHT.

4. Project Workflow

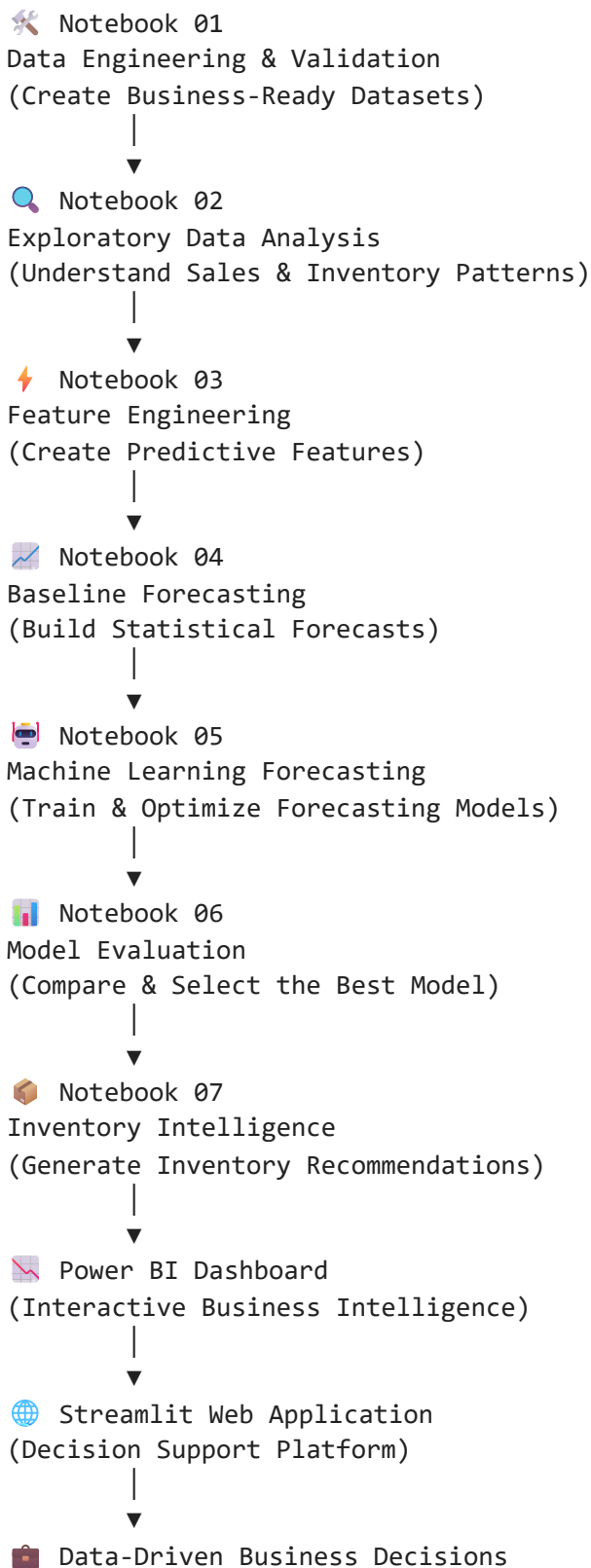
Notebook 01 is the starting point of **Project FORESIGHT**. It transforms the raw M5 datasets into clean, validated, and business-ready datasets that serve as the foundation for all subsequent stages of the project.

End-to-End Project Pipeline

Project FORESIGHT

Raw M5 Retail Datasets





Deliverables of Notebook 01

By the end of this notebook, the following business-ready datasets will be created:

-  Sales Daily Dataset
-  SKU Master Dataset
-  Enhanced Calendar Dataset
-  Inventory Snapshot Dataset

Additionally, all datasets will be:

- Clean
- Validated
- Business-ready
- Ready for analytics
- Ready for machine learning
- Ready for dashboard development

Outputs Generated

Upon successful completion of Notebook 01, the following validated datasets will be available:

Output	Description
<code>sales_daily_final.csv</code>	Daily SKU-level sales dataset prepared for analysis and forecasting
<code>calendar_final.csv</code>	Enhanced Calendar dimension containing engineered temporal, event, and SNAP attributes
<code>sku_master_final.csv</code>	Canonical SKU-store reference containing hierarchy, lifecycle, pricing, and estimated financial attributes
<code>inventory_snapshots_final.csv</code>	Simulated daily inventory dataset containing stock movements, replenishment activity, status, and inventory metrics

The datasets are stored in the `data/processed/` directory and serve as the validated inputs for subsequent Project FORESIGHT notebooks.

Key Takeaway

Notebook 01 establishes the validated analytical data foundation of Project FORESIGHT. The quality and reliability of every subsequent analysis, forecasting model, dashboard, and business recommendation depend on the datasets engineered, validated, and exported in this notebook.

5. Project Configuration

This section prepares the notebook environment to ensure a clean, consistent, and reproducible workflow throughout **Project FORESIGHT**.

The configuration performed in this section includes:

- Suppressing unnecessary warning messages
- Establishing notebook-wide settings
- Preparing the environment before importing project libraries

A properly configured notebook improves readability, minimizes distractions during analysis, and provides a consistent development experience.

✔ Notebook environment configured successfully.

Configuration Summary

The notebook environment has now been configured successfully.

In this step, warning messages are suppressed to keep the notebook output clean and focused during data exploration and feature engineering. This improves readability while working with large datasets and multiple libraries.

Note: Warning suppression does not resolve underlying issues. During development, important warnings should always be reviewed before deciding whether to suppress them.

6. Import Required Libraries

This section imports and configures the Python libraries required for data manipulation, numerical operations, file-path management, and notebook display.

The libraries are organized by purpose to maintain a clear and consistent notebook structure.

✔ Section 6 completed: Required libraries imported successfully.

Library Summary

The required Python libraries were successfully imported and configured for Notebook 01. Each library supports a specific aspect of the data engineering workflow.

Library	Purpose
pandas	Data manipulation, cleaning, and transformation
NumPy	Numerical computations and array operations
Pathlib	File and directory management
IPython.display	Enhanced display of notebook outputs

The notebook environment is now fully configured for the subsequent stages of Project FORESIGHT.

7. Project Directory & File Paths

This section defines reusable project paths for source datasets and engineered outputs, ensuring consistent file access throughout Notebook 01.

This section will:

- identify the project root and data directories;
- define the source paths for the Sales, Calendar, and Selling Prices datasets;

- define the processed-data location for engineered outputs.

Using `pathlib.Path` keeps file-path handling readable and portable across operating systems.

✅ Section 7 completed: Project directories and file paths configured successfully.

Directory Summary

The project directory structure has been successfully configured to support a consistent and organized data engineering workflow throughout Notebook 01.

Directory	Purpose
data/raw/	Stores the original M5 datasets used as notebook inputs.
data/processed/	Stores the engineered datasets generated by Notebook 01 for downstream analytics and forecasting.
data/external/	Reserved for additional external datasets that may be integrated into future stages of Project FORESIGHT.

The project directory structure is now ready to support the remaining stages of the Project FORESIGHT pipeline.

 **Key Takeaway:** Organizing datasets into dedicated directories for raw, processed, and external data improves project maintainability, reproducibility, and scalability by ensuring a clear separation between original and engineered data assets.

8. Load Raw M5 Datasets

After configuring the project directories and file paths, the three original M5 datasets are loaded into Pandas DataFrames. These datasets provide the raw inputs required for exploration, quality assessment, and data engineering.

This section will:

- load the historical sales dataset;
- load the calendar and selling-prices datasets;
- confirm that all source datasets were loaded successfully.

 **Note:** No cleaning or transformation is performed at this stage. The datasets are loaded in their original form for subsequent assessment.

✅ All raw M5 datasets loaded successfully.

Sales dataset shape : (30490, 1919)

Calendar dataset shape: (1969, 14)

Prices dataset shape : (6841121, 4)

Dataset Loading Summary

The three raw M5 datasets were successfully loaded into Pandas DataFrames using the configured source file paths.

DataFrame	Source Dataset	Purpose
<code>sales_df</code>	<code>sales_train_validation.csv</code>	Contains historical daily unit sales across products and stores.
<code>calendar_df</code>	<code>calendar.csv</code>	Contains dates, events, calendar attributes, and SNAP indicators.
<code>prices_df</code>	<code>sell_prices.csv</code>	Contains weekly selling prices for item-store combinations.

💡 **Key Takeaway:** The datasets remain unchanged from their original source files at this stage. Cleaning, transformation, and feature engineering will be performed only after completing the initial exploration and quality assessment.

9. Source Data Understanding

Understanding the raw datasets is the first step in the analytics workflow. Before assessing data quality or performing data engineering, it is essential to understand each dataset's structure, business purpose, and relationships.

In this section, we will:

- Explore the Sales dataset.
- Explore the Calendar dataset.
- Explore the Selling Prices dataset.
- Examine how the datasets are connected.

This understanding provides the foundation for the data quality assessment and engineering tasks in the following sections.

✳️ **Note:** This section focuses only on understanding the raw datasets. No cleaning, validation, or transformation is performed.

9.1 Sales Dataset Overview

The Sales dataset contains historical daily demand for 30,490 item-store combinations.

Each row represents one item-store series, while columns `d_1` through `d_1913` contain daily units sold. This section reviews the dataset's dimensions, schema, identifiers, and business role before quality assessment.

In this subsection, we will:

- Preview the dataset.
- Examine its dimensions.

- Review its schema.
- Understand its business context.

9.1.1 Dataset Preview & Dimensions

Previewing the dataset and examining its dimensions provide an initial understanding of its structure and scale. These checks verify that the data has been loaded correctly and establish the foundation for subsequent schema inspection.

	id	item_id	dept_id	cat_id	store_id	state_id	d_1	d_2
0	HOBBIES_1_001_CA_1_validation	HOBBIES_1_001	HOBBIES_1	HOBBIES	CA_1	CA	0	0
1	HOBBIES_1_002_CA_1_validation	HOBBIES_1_002	HOBBIES_1	HOBBIES	CA_1	CA	0	0
2	HOBBIES_1_003_CA_1_validation	HOBBIES_1_003	HOBBIES_1	HOBBIES	CA_1	CA	0	0
3	HOBBIES_1_004_CA_1_validation	HOBBIES_1_004	HOBBIES_1	HOBBIES	CA_1	CA	0	0
4	HOBBIES_1_005_CA_1_validation	HOBBIES_1_005	HOBBIES_1	HOBBIES	CA_1	CA	0	0

=====

Number of Rows : 30,490
Number of Columns : 1919

=====

✔ Sales dataset preview and dimensions displayed successfully.

9.1.2 Schema Inspection (Columns & Data Types)

Schema inspection provides an overview of the dataset's columns, data types, non-null values, and memory usage. This helps verify the dataset structure and understand how the information is organized before further analysis.

 **Note:** This subsection inspects the dataset schema only. Data quality assessment and transformations are performed in later sections.

Column Names:

```
Index(['id', 'item_id', 'dept_id', 'cat_id', 'store_id', 'state_id', 'd_1', 'd_2', 'd_3',
      'd_4',
      ...,
      'd_1904', 'd_1905', 'd_1906', 'd_1907', 'd_1908', 'd_1909', 'd_1910', 'd_1911', 'd_
1912', 'd_1913'],
      dtype='str', length=1919)
```

```
<class 'pandas.DataFrame'>
RangeIndex: 30490 entries, 0 to 30489
Columns: 1919 entries, id to d_1913
dtypes: int64(1913), str(6)
memory usage: 448.2 MB
```

✅ Sales dataset schema displayed successfully.

9.1.3 Business Context

The Sales dataset captures the historical demand for products sold across multiple stores. Each row represents a unique product-store combination, while the daily columns (`d_1` to `d_1913`) record the number of units sold each day.

The first six columns describe the product and store:

- **id** – Unique identifier for each product-store combination.
- **item_id** – Product identifier.
- **dept_id** – Product department.
- **cat_id** – Product category.
- **store_id** – Store identifier.
- **state_id** – State identifier.

The remaining **1,913** columns contain the daily sales history, forming a wide-format time-series dataset that will later be transformed into a long-format dataset for feature engineering and forecasting.

📌 **Business Importance:** Historical sales data enables demand forecasting, inventory optimization, and data-driven supply chain planning.

9.1.4 Key Observations

The initial exploration reveals the following:

- The dataset contains **30,490** product-store combinations.
- It consists of **1,919 columns**, including **6 identifier columns** and **1,913 daily sales columns** (`d_1` – `d_1913`).
- Identifier columns use the **str** data type, while daily sales values are stored as **int64** .
- The dataset follows a **wide-format time-series structure**.

- It serves as the primary source for demand forecasting and will later be reshaped into a long-format dataset for analysis and feature engineering.

Sales Dataset Summary

The Sales dataset is the primary source of historical demand information in the M5 Forecasting project. Understanding its structure, schema, and business context establishes the foundation for data quality assessment, feature engineering, and demand forecasting.

 **Key Takeaway:** A clear understanding of the raw dataset is essential before performing any validation, transformation, or analysis.

9.2 Calendar Dataset Overview

The Calendar dataset maps encoded sales days to actual dates and provides temporal, event, and SNAP information.

This section reviews its dimensions, schema, key fields, and role in explaining seasonality and other time-based demand patterns.

In this subsection, we will:

- Preview the dataset.
- Examine its dimensions.
- Review its schema.
- Understand its business context.

9.2.1 Dataset Preview & Dimensions

Previewing the dataset and examining its dimensions provide an initial understanding of its structure and scale. These checks verify that the data has been loaded correctly and prepare us for schema inspection.

	date	wm_yr_wk	weekday	wday	month	year	d	event_name_1	event_type_1	event_name_2
0	2011-01-29	11101	Saturday	1	1	2011	d_1	NaN	NaN	
1	2011-01-30	11101	Sunday	2	1	2011	d_2	NaN	NaN	
2	2011-01-31	11101	Monday	3	1	2011	d_3	NaN	NaN	
3	2011-02-01	11101	Tuesday	4	2	2011	d_4	NaN	NaN	
4	2011-02-02	11101	Wednesday	5	2	2011	d_5	NaN	NaN	

```
=====
Number of Rows      : 1,969
Number of Columns   : 14
=====
```

✅ Calendar dataset preview and dimensions displayed successfully.

9.2.2 Schema Inspection (Columns & Data Types)

Schema inspection provides an overview of the dataset's columns, data types, non-null values, and memory usage. This helps verify the dataset structure and understand how the calendar information is organized before further analysis.

✳️ **Note:** This subsection inspects the dataset schema only. Data quality assessment and transformations are performed in later sections.

Column Names:

```
Index(['date', 'wm_yr_wk', 'weekday', 'wday', 'month', 'year', 'd', 'event_name_1', 'event_type_1', 'event_name_2', 'event_type_2', 'snap_CA', 'snap_TX', 'snap_WI'],
      dtype='str')
```

```
=====
<class 'pandas.DataFrame'>
RangeIndex: 1969 entries, 0 to 1968
Data columns (total 14 columns):
#   Column          Non-Null Count  Dtype
---  -
0   date             1969 non-null   str
1   wm_yr_wk         1969 non-null   int64
2   weekday          1969 non-null   str
3   wday             1969 non-null   int64
4   month            1969 non-null   int64
5   year             1969 non-null   int64
6   d               1969 non-null   str
7   event_name_1     162 non-null    str
8   event_type_1     162 non-null    str
9   event_name_2     5 non-null      str
10  event_type_2     5 non-null      str
11  snap_CA          1969 non-null   int64
12  snap_TX          1969 non-null   int64
13  snap_WI          1969 non-null   int64
dtypes: int64(7), str(7)
memory usage: 263.1 KB
=====
```

✅ Calendar dataset schema displayed successfully.

9.2.3 Business Context

The Calendar dataset provides the temporal context for historical sales data. While the Sales dataset records **what** was sold, the Calendar dataset explains **when** the sales occurred and identifies external factors that may influence demand.

The dataset can be understood through three logical groups:

1. Calendar Attributes

The calendar-related columns (`date` , `wm_yr_wk` , `weekday` , `wday` , `month` , `year` , and `d`) map encoded day identifiers to actual calendar dates, enabling the extraction of time-based features.

2. Event Information

The event-related columns (`event_name_1` , `event_type_1` , `event_name_2` , and `event_type_2`) capture holidays, cultural celebrations, sporting events, and other special occasions that may influence customer purchasing behavior.

3. SNAP Indicators

The SNAP columns (`snap_CA` , `snap_TX` , and `snap_WI`) indicate whether the Supplemental Nutrition Assistance Program (SNAP) was active in California, Texas, and Wisconsin, providing additional socioeconomic context for demand forecasting.

✦ **Business Importance:** Calendar information enables forecasting models to capture seasonality, holiday effects, and other time-based demand patterns.

9.2.4 Key Observations

The initial exploration reveals the following:

- The dataset contains **1,969 records** and **14 columns**.
- It consists of **7 str columns** and **7 int64 columns**.
- The encoded day identifier (`d`) links the Calendar dataset with the Sales dataset.
- The dataset captures calendar attributes, events, and SNAP indicators that provide valuable context for demand forecasting.
- It serves as the temporal backbone of the forecasting pipeline and supports the creation of time-based features.



Calendar Dataset Summary

The Calendar dataset provides the temporal and contextual information required to interpret historical sales data. Understanding its structure and business context establishes the foundation for time-based feature engineering and demand forecasting.

💡 **Key Takeaway:** Calendar information enriches historical sales data with seasonality, events, and socioeconomic context, enabling more accurate demand forecasting.



9.3 Selling Prices Dataset Overview

The Selling Prices dataset contains weekly item-store pricing records aligned through the retail-week identifier `wm_yr_wk`.

This section reviews its dimensions, schema, business keys, and role in pricing analysis and demand forecasting.

9.3.1 Dataset Preview & Dimensions

Previewing the dataset and examining its dimensions provide an initial understanding of its structure and scale. These checks verify that the data has been loaded correctly and prepare us for schema inspection.

	store_id	item_id	wm_yr_wk	sell_price
0	CA_1	HOBBIES_1_001	11325	9.58
1	CA_1	HOBBIES_1_001	11326	9.58
2	CA_1	HOBBIES_1_001	11327	8.26
3	CA_1	HOBBIES_1_001	11328	8.26
4	CA_1	HOBBIES_1_001	11329	8.26

=====

Number of Rows : 6,841,121
Number of Columns : 4

=====

✅ Selling Prices dataset preview and dimensions displayed successfully.

9.3.2 Schema Inspection (Columns & Data Types)

Schema inspection provides an overview of the dataset's columns, data types, non-null values, and memory usage. This helps verify the dataset structure and understand how pricing information is organized before further analysis.

✳️ **Note:** This subsection inspects the dataset schema only. Data quality assessment and transformations are performed in later sections.

Column Names:

```
Index(['store_id', 'item_id', 'wm_yr_wk', 'sell_price'], dtype='str')
```

```
=====

<class 'pandas.DataFrame'>
RangeIndex: 6841121 entries, 0 to 6841120
Data columns (total 4 columns):
#   Column      Dtype
---  -
0   store_id    str
1   item_id     str
2   wm_yr_wk    int64
3   sell_price  float64
dtypes: float64(1), int64(1), str(2)
memory usage: 318.1 MB

=====
```

✅ Selling Prices dataset schema displayed successfully.

9.3.3 Business Context

The Selling Prices dataset records the weekly selling price of each product across different stores. While the Sales dataset captures customer demand, this dataset provides the pricing information needed to analyze the impact of price changes on demand.

The dataset is organized around four key variables:

- **store_id** – Store identifier.
- **item_id** – Product identifier.
- **wm_yr_wk** – Retail week used to align pricing information with the Calendar dataset.
- **sell_price** – Weekly selling price of the product.

🔴 **Business Importance:** Pricing information enables forecasting models to capture the impact of price fluctuations, improving demand forecasting and inventory planning.

9.3.4 Key Observations

The initial exploration reveals the following:

- The dataset contains **6,841,121 records** and **4 columns**, making it the largest of the three M5 source datasets.
- It consists of **2 str columns**, **1 int64 column**, and **1 float64 column**.
- Each record represents the weekly selling price of a product in a specific store.
- The combination of **store_id**, **item_id**, and **wm_yr_wk** uniquely identifies a pricing record.
- The dataset provides the pricing context required to analyze the relationship between price changes and customer demand.

\$ Selling Prices Dataset Summary

The Selling Prices dataset provides the pricing dimension of the M5 Forecasting project. Understanding its structure, schema, and business context establishes the foundation for pricing analysis and demand forecasting.

💡 **Key Takeaway:** Pricing data complements historical sales and calendar information, enabling forecasting models to capture the impact of price changes on customer demand.

🔗 9.4 Dataset Relationship Overview

The three source datasets combine demand, calendar context, and weekly pricing into a unified analytical structure.

9.4.1 Dataset Roles

- The **Sales dataset** records daily item-store demand.
- The **Calendar dataset** maps encoded sales days to dates, retail weeks, events, and SNAP indicators.
- The **Selling Prices dataset** provides weekly item-store prices.

Together, they explain what was sold, when it was sold, and at what price.

9.4.2 Common Keys and Join Strategy

The datasets are connected through the following fields:

- `d` links reshaped daily sales records with the Calendar dataset.
- `wm_yr_wk` connects calendar dates with weekly pricing records.
- `item_id` and `store_id` identify the applicable item-store price.

The Sales dataset will first be reshaped to long format, joined with Calendar using `d`, and then enriched with pricing information using `item_id`, `store_id`, and `wm_yr_wk`.

9.4.3 Integration Workflow

1. Reshape the Sales dataset into one record per item-store-day.
2. Attach calendar dates, events, and SNAP attributes using `d`.
3. Attach weekly selling prices using `item_id`, `store_id`, and `wm_yr_wk`.
4. Use the integrated records for feature engineering, forecasting, and inventory analysis.

🌟 **Business Importance:** Integrating these datasets enables forecasting models to capture historical demand, seasonality, events, and pricing behavior, supporting more accurate demand forecasting and inventory planning.

Source Data Understanding Summary

In this section, we explored the three M5 source datasets and their roles in the demand forecasting pipeline. We examined their structure, schema, business context, and relationships to establish a clear understanding of the data before moving to the engineering phase.

 **Next Step:** Section 10 focuses on Data Quality Assessment, where we will evaluate missing values, duplicates, inconsistencies, and other quality issues before data integration and transformation.

10. Data Quality Assessment

This section evaluates the completeness, uniqueness, data types, key integrity, and business-rule validity of the three source datasets before data engineering begins.

This section will:

- assess the Sales, Calendar, and Selling Prices datasets independently;
- distinguish genuine data-quality issues from valid business conditions;
- confirm each dataset's readiness for transformation and integration.

No cleaning or feature engineering is performed in this section.

10.1 Sales Dataset Quality Assessment

This subsection evaluates the Sales dataset for:

- missing values;
- duplicate records;
- data-type consistency;
- identifier integrity;
- valid non-negative sales values.

The results will determine whether the dataset is ready for transformation in Section 11.

10.1.1 Missing Values

Missing values can reduce data quality and impact downstream analysis, feature engineering, and predictive modeling. This subsection evaluates the completeness of the Sales dataset by identifying missing values across all columns.

 **Objective:** Verify whether the Sales dataset contains any missing values that require attention before data engineering.

```
Series([], dtype: int64)
```


=====

Total Missing Values : 0
Columns with Missing Values : 0

=====

✅ No missing values detected in the Sales dataset.

Quality Assessment

No missing values were detected across the 1,919 Sales dataset columns.

All identifier fields and 1,913 daily sales columns are complete, confirming that the dataset can proceed to the remaining quality checks without missing-data treatment.

10.1.2 Duplicate Records

Duplicate records can introduce bias, inflate analytical results, and affect data integrity. This subsection evaluates whether the Sales dataset contains duplicate rows that require further investigation.

✦ **Objective:** Identify duplicate records and determine whether they represent genuine data quality issues.

Duplicate Records : 0

=====

✅ No duplicate records detected in the Sales dataset.

=====

✅ Duplicate record assessment completed.

Quality Assessment

No duplicate rows were detected in the Sales dataset.

Each of the 30,490 records represents one unique product-store sales series, preventing duplicate demand histories and double-counting during downstream processing.

10.1.3 Data Type Validation

Data type validation verifies that each column is stored using an appropriate data type. Correct data types improve computational efficiency, reduce processing errors, and ensure compatibility with downstream data engineering and analytical tasks.

✦ **Objective:** Validate the data types of all columns in the Sales dataset and identify any inconsistencies that may require correction.

```
int64      1913
str         6
Name: count, dtype: int64
=====
```

	Column	Data Type
0	id	str
1	item_id	str
2	dept_id	str
3	cat_id	str
4	store_id	str
...	...	...
1914	d_1909	int64
1915	d_1910	int64
1916	d_1911	int64
1917	d_1912	int64
1918	d_1913	int64

1919 rows × 2 columns

=====

✔ Data type validation completed.

Quality Assessment

The Sales dataset contains 1,913 integer-based daily sales columns and six string-based identifier columns.

No unexpected data types were detected. The schema is consistent with the dataset’s wide-format structure and is suitable for subsequent reshaping and engineering.

10.1.4 Identifier Validation

Identifier columns uniquely describe each product-store combination and play a critical role in data integration and feature engineering. This subsection validates the identifier columns by checking for missing values, uniqueness, and cardinality.

 **Objective:** Verify that the identifier columns are complete and suitable for downstream data integration.

	Unique Values	Missing Values
id	30490	0
item_id	3049	0
dept_id	7	0
cat_id	3	0
store_id	10	0
state_id	3	0

=====

Duplicate 'id' Values : 0

=====

- ✓ No missing values detected in the identifier columns.
- ✓ The 'id' column uniquely identifies each product-store combination.

=====

- ✓ Identifier validation completed.

Quality Assessment

All identifier fields are complete, and the **id** column uniquely identifies all 30,490 product-store sales series.

The dataset contains 3,049 items, seven departments, three categories, 10 stores, and three states. No identifier-level integrity issues were detected.

10.1.5 Sales Value Validation

Sales value validation verifies that daily demand values are stored in a valid and consistent form. Since sales represent units sold, values should be non-negative integers, while zero values may legitimately indicate days with no demand.

- ✦ **Objective:** Identify invalid sales values and confirm that the daily sales columns are suitable for downstream forecasting.

	Metric	Result
0	Sales Columns	1913
1	Total Sales Values	58327370
2	Minimum Sales Value	0
3	Maximum Sales Value	763
4	Negative Values	0
5	Non-Integer Values	0
6	Zero Sales Values	39777094

=====

✓ Daily sales values are valid non-negative integers.

=====

✓ Sales value validation completed.

Quality Assessment

The 1,913 daily sales columns contain 58,327,370 non-negative integer observations ranging from 0 to 763 units.

No negative or non-integer values were detected. The 39,777,094 zero values represent valid no-demand observations rather than missing data.

✓ Sales Dataset Quality Assessment Summary

The Sales dataset passed all assessed quality checks.

No missing values, duplicate records, identifier inconsistencies, negative sales values, or non-integer demand observations were detected. Valid zero-demand records were retained as meaningful historical observations.

The dataset is ready for transformation and integration in Section 11.

10.2 Calendar Dataset Quality Assessment

This subsection evaluates the Calendar dataset for:

- missing values;
- duplicate records;
- data-type consistency;
- calendar-key integrity;
- valid event and SNAP attributes.

The assessment will distinguish expected business-related missing values from genuine data-quality issues.

10.2.1 Missing Values

Missing values can indicate incomplete data or, in some cases, represent valid business conditions. This subsection evaluates the completeness of the Calendar dataset and distinguishes between expected and unexpected missing values.

🔴 **Objective:** Identify missing values and determine whether they represent genuine data quality issues or valid business scenarios.

	Missing Values	Percentage (%)
event_name_1	1807	91.77
event_type_1	1807	91.77
event_name_2	1964	99.75
event_type_2	1964	99.75

=====
Total Missing Values : 7,542
Columns with Missing Values : 4
=====

✅ Calendar missing value assessment completed.

Quality Assessment

Missing values occur only in the four event-related columns.

The matching missing counts between each event name and event type confirm that these values represent dates without scheduled events rather than incomplete records. All remaining Calendar and SNAP fields are complete.

10.2.2 Duplicate Records

Duplicate records can introduce redundancy, affect data integrity, and lead to incorrect results during dataset integration and time-based analysis. This subsection evaluates whether the Calendar dataset contains duplicate rows that require further investigation.

🔴 **Objective:** Verify that each calendar record represents a unique date and identify any duplicate records that may affect downstream data engineering.

Duplicate Records : 0

=====

✓ No duplicate records detected in the Calendar dataset.

=====

✓ Duplicate record assessment completed.

Quality Assessment

No duplicate rows were detected in the Calendar dataset.

Each of the 1,969 records represents one unique calendar date, supporting unambiguous date-based joins and chronological analysis.

10.2.3 Data Type Validation

Data type validation verifies that each column is stored using an appropriate data type. Correct data types improve data consistency, reduce processing errors, and ensure compatibility with downstream data engineering and analytical tasks.

✦ **Objective:** Validate the data types of all columns in the Calendar dataset and identify any inconsistencies that may require correction.

```
str      7
int64    7
Name: count, dtype: int64
```

=====

	Column	Data Type
0	date	str
1	wm_yr_wk	int64
2	weekday	str
3	wday	int64
4	month	int64
5	year	int64
6	d	str
7	event_name_1	str
8	event_type_1	str
9	event_name_2	str
10	event_type_2	str
11	snap_CA	int64
12	snap_TX	int64
13	snap_WI	int64

=====

✅ Calendar data type validation completed.

Quality Assessment

The Calendar dataset contains seven string columns and seven integer columns.

All source data types are appropriate for initial assessment. The `date` field remains string-based and will be converted to `datetime` during calendar engineering; no other unexpected data-type issues were detected.

10.2.4 Calendar Attribute Validation

Calendar attribute validation evaluates the completeness, uniqueness, and cardinality of the core temporal fields. These checks confirm that the Calendar dataset contains one record per date and one unique encoded day identifier before downstream data engineering.

✦ **Objective:** Verify that the core calendar attributes are complete and that the `date` and `d` columns uniquely identify each calendar record.

	Unique Values	Missing Values
date	1969	0
wm_yr_wk	282	0
weekday	7	0
wday	7	0
month	12	0
year	6	0
d	1969	0

Duplicate 'd' Values : 0

Duplicate Dates : 0

- ✓ No missing values detected in the core calendar attributes.
- ✓ Calendar day identifiers and dates are unique.

- ✓ Calendar attribute validation completed.

Quality Assessment

All core Calendar attributes are complete.

The dataset contains 1,969 unique dates and 1,969 unique **d** identifiers, with no duplicate key values. The remaining attributes cover 282 retail weeks, 12 months, six years, and seven weekday categories.

10.2.5 Event Information Validation

The Calendar dataset contains event-related attributes that capture holidays, cultural celebrations, sporting events, and other special occasions that may influence customer demand. This subsection validates the completeness and consistency of the event information to ensure it is suitable for downstream feature engineering and demand forecasting.

✦ **Objective:** Validate the event-related columns and confirm that event information is logically consistent and suitable for downstream analytical tasks.

	Unique Values	Missing Values
event_name_1	30	1807
event_type_1	4	1807
event_name_2	4	1964
event_type_2	2	1964

=====

Primary Event Mismatches : 0
 Secondary Event Mismatches : 0

=====

✔ Event names and event types are logically consistent.

=====

✔ Event information validation completed.

Quality Assessment

The Calendar dataset contains 30 primary event names across four event types and four secondary event names across two event types.

No mismatches were detected between event names and event types. Missing values represent dates without scheduled events and do not indicate incomplete records.

10.2.6 SNAP Indicator Validation

The SNAP (Supplemental Nutrition Assistance Program) indicators identify whether SNAP benefits are available on a given day in California (CA), Texas (TX), and Wisconsin (WI). These binary variables are important external factors that may influence customer purchasing behavior and demand patterns. This subsection validates the SNAP indicators to ensure they are complete, consistent, and contain only valid binary values.

✦ **Objective:** Validate the SNAP indicator columns and confirm that they contain complete and valid binary values suitable for downstream feature engineering and demand forecasting.

	Unique Values	Missing Values	Minimum	Maximum
snap_CA	2	0	0	1
snap_TX	2	0	0	1
snap_WI	2	0	0	1

=====

Invalid Values	
snap_CA	0
snap_TX	0
snap_WI	0

=====

✓ All SNAP indicators contain valid binary values (0 or 1).

=====

✓ SNAP indicator validation completed.

Quality Assessment

The `snap_CA` , `snap_TX` , and `snap_WI` fields are complete and contain only the valid binary values `0` and `1` .

No missing or invalid SNAP indicator values were detected, confirming that the fields are ready for Calendar feature engineering.

✓ Calendar Dataset Quality Assessment Summary

The Calendar dataset passed all assessed quality checks.

Core calendar keys are complete and unique, event-field missing values represent valid non-event dates, event names and types are logically aligned, and all SNAP indicators contain valid binary values.

The `date` field requires conversion to `datetime` during Section 11; no other corrective treatment is required.

\$ 10.3 Selling Prices Dataset Quality Assessment

This subsection evaluates the Selling Prices dataset for:

- missing values;
- duplicate records;
- data-type consistency;
- composite-key integrity;
- valid positive selling prices.

The results will determine whether the weekly pricing records are ready for integration with Sales and Calendar.

10.3.1 Missing Values

Missing values can reduce data quality and impact price analysis, feature engineering, and demand forecasting. This subsection evaluates the completeness of the Selling Prices dataset by identifying missing values across all columns.

✦ **Objective:** Verify whether the Selling Prices dataset contains any missing values that require attention before data engineering.

```
Series([], dtype: int64)
=====
```

```
Total Missing Values : 0
Columns with Missing Values : 0
```

```
=====
```

✓ No missing values detected in the Selling Prices dataset.

```
=====
```

✓ Selling Prices missing value assessment completed.

Quality Assessment

No missing values were detected across the four Selling Prices fields.

All product, store, retail-week, and selling-price records are complete, so no missing-data treatment is required before pricing integration.

10.3.2 Duplicate Records

Duplicate records can introduce redundancy, distort price analysis, and affect data integrity during dataset integration. This subsection evaluates whether the Selling Prices dataset contains duplicate records that require further investigation.

✦ **Objective:** Identify duplicate records and determine whether they represent genuine data quality issues.

```
Duplicate Records : 0
=====
```

✓ No duplicate records detected in the Selling Prices dataset.

```
=====
```

✓ Selling Prices duplicate record assessment completed.

Quality Assessment

No duplicate rows were detected in the Selling Prices dataset.

Each record represents one unique store-item-week pricing observation, preventing duplicate price assignments during integration.

10.3.3 Data Type Validation

Data type validation verifies that each column is stored using an appropriate data type. Correct data types improve computational efficiency, reduce processing errors, and ensure compatibility with downstream data engineering and analytical tasks.

✦ **Objective:** Validate the data types of all columns in the Selling Prices dataset and identify any inconsistencies that may require correction.

```
str          2
int64        1
float64       1
Name: count, dtype: int64
```

=====

	Column	Data Type
0	store_id	str
1	item_id	str
2	wm_yr_wk	int64
3	sell_price	float64

=====

✅ Selling Prices data type validation completed.

Quality Assessment

The Selling Prices dataset contains two string identifier fields, one integer retail-week field, and one floating-point selling-price field.

No unexpected or inconsistent data types were detected. The schema is appropriate for pricing integration and downstream calculations.

10.3.4 Identifier Validation

Identifier columns uniquely define each product-store-week combination and play a critical role in data integration and feature engineering. This subsection validates the identifier columns by checking for missing values, uniqueness, and cardinality.

✦ **Objective:** Verify that the identifier columns are complete and suitable for downstream data integration with the Sales and Calendar datasets.

	Unique Values	Missing Values
store_id	10	0
item_id	3049	0
wm_yr_wk	282	0

=====

Duplicate Product-Store-Week Combinations : 0

=====

- ✔ No missing values detected in the identifier columns.
- ✔ Each product-store-week combination is unique.

=====

- ✔ Selling Prices identifier validation completed.

Quality Assessment

All identifier fields are complete.

The dataset contains 10 stores, 3,049 items, and 282 retail weeks. Every `store_id – item_id – wm_yr_wk` combination is unique, confirming a valid composite key for pricing integration.

10.3.5 Selling Price Validation

Selling price is the primary business metric in this dataset and directly influences revenue analysis, pricing strategy, and demand forecasting. This subsection validates the selling price values by identifying invalid, missing, zero, or negative prices and summarizing the overall price distribution.

📌 **Objective:** Assess the completeness and validity of selling price values to ensure they are suitable for downstream analytical and forecasting tasks.

	Metric	Value
0	Minimum Selling Price	0.01
1	Maximum Selling Price	107.32
2	Average Selling Price	4.41
3	Missing Prices	0.00
4	Zero Prices	0.00
5	Negative Prices	0.00

=====

✅ Selling price values passed the validation checks.

=====

✅ Selling Price validation completed.

Quality Assessment

All selling prices are complete, numeric, and strictly positive.

Observed prices range from 0.01 to 107.32, with an average of 4.41. No missing, zero, or negative price values were detected.

✅ Selling Prices Quality Assessment Summary

The Selling Prices dataset passed all assessed quality checks.

No missing values, duplicate records, data-type inconsistencies, composite-key violations, or invalid selling prices were detected.

The dataset is ready for integration with the Sales and Calendar datasets in Section 11.

Section 10 Summary

The Sales, Calendar, and Selling Prices datasets completed all required quality assessments.

The source datasets contain valid identifiers, appropriate data types, consistent business keys, and no unresolved quality issues. Expected event-field missing values were retained as meaningful non-event dates.

The datasets are ready for engineering and integration in Section 11.

 **Next Step:** Section 11 focuses on Data Engineering, where we will clean, transform, and integrate the validated datasets to create business-ready data structures for exploratory data analysis, feature engineering, and demand forecasting.

11. Data Engineering

This section transforms the validated source data into four standardized analytical datasets:

- Sales Daily Dataset;
- Calendar Dataset;
- SKU Master Dataset;
- simulated Inventory Snapshot Dataset.

Each dataset will be engineered and validated independently before final cross-dataset validation and export.

11.1 Sales Daily Dataset Engineering

The raw sales dataset is transformed into a daily transactional format by integrating calendar and pricing information and engineering business-ready features.

Objective:

- Prepare the sales dataset for analysis and forecasting.
- Engineer business-relevant features.
- Create a clean and standardized daily sales dataset.

11.1.1 Create Sales Working Copy

A working copy of the raw sales dataset is created before applying any transformations. This preserves the original dataset and ensures a safe, reproducible data engineering workflow.

Objective:

- Preserve the raw dataset.
- Create an independent working copy.
- Perform all transformations safely.

✅ Sales Daily working dataset created successfully.

Dataset Shape: (30490, 1919)

Independent Working Copy: True

✅ Validation Result

A separate working copy of the Sales dataset was created successfully.

Validation confirms that:

- the working dataset preserves the original dimensions;
- the working copy is independent of the raw Sales dataset;
- all subsequent transformations will be performed safely without modifying the original source data.

11.1.2 Identify Sales-Day Columns

The sales dataset stores daily sales values across multiple day columns (`d_1` to `d_n`). These columns are identified dynamically to support the transformation from wide format to long format.

Objective:

- Identify all daily sales columns.
- Separate identifier and sales columns.
- Prepare the dataset for reshaping.

Sales-day columns identified: 1913

First 5 sales-day columns:

```
['d_1', 'd_2', 'd_3', 'd_4', 'd_5']
```

Last 5 sales-day columns:

```
['d_1909', 'd_1910', 'd_1911', 'd_1912', 'd_1913']
```

Identifier columns defined successfully.

```
['id', 'item_id', 'dept_id', 'cat_id', 'store_id', 'state_id']
```

All identifier columns are present in the sales dataset.

✅ Validation Result

The daily sales columns were identified successfully.

Validation confirms that:

- all day-level sales columns were detected dynamically;
- the identified columns are ready for reshaping into long format;
- no manual column selection is required for subsequent transformations.

⚙️ 11.1.3 Configure the Development Sales Scope

The complete M5 sales dataset contains more than 30,000 product-store time series and would generate over 58 million records after conversion from wide to long format. Processing the complete dataset may require substantial memory, storage, and execution time on a local system.

To support efficient and reproducible local development, a category-balanced subset of 30 products—10 from each category—was selected using deterministic sorting, while retaining all 10 stores and all 1,913 historical sales days.

This subset supports pipeline development across product categories, stores, states and time periods, but it should not be interpreted as a statistically representative sample of the complete product or department population.

Objective:

- Define a reproducible and laptop-friendly sales development scope.
- Select an equal and reproducible number of development products from each product category.
- Retain all available stores and historical sales days.
- Reduce computational and storage requirements for downstream processing.

Development Scope Configuration

Products selected per category : 10

Available product categories : 3

Available retail stores : 10

Historical sales-day columns : 1913

	cat_id	selected_products
0	FOODS	10
1	HOBBIES	10
2	HOUSEHOLD	10

Development dataset created: (300, 1919)

✅ Development sales dataset validated successfully.

Unique products : 30

Unique retail stores : 10

Unique SKU-store series : 300

Expected long-format rows: 573,900

✅ Validation Result

The development sales scope was configured successfully and validated against the expected engineering requirements.

Validation confirmed that:

30 category-balanced development products were selected, with 10 products retained from each of the 3 product categories.

- All 10 retail stores were retained in the development dataset.
- The dataset contains 300 unique SKU-store series.
- All 1,913 historical sales-day columns were preserved.
- The expected long-format dataset will contain 573,900 daily sales records after reshaping.

The validated development dataset provides a category-balanced, reproducible, and computationally efficient foundation for developing the remaining sales engineering pipeline within the defined 300-series scope.

11.1.4 Reshape the Sales Dataset to Long Format

The development sales dataset currently stores each historical sales day in a separate column. Although this wide structure is suitable for compact storage, it is not convenient for time-series analysis, relational joins, feature engineering, or downstream analytical workflows.

The dataset is therefore converted into long format, where each row represents the units sold for one SKU-store series on one historical sales day. This structure will support subsequent integration with calendar and pricing data.

Objective:

- Convert the wide sales dataset into a row-level daily sales structure.
- Preserve all product, store, category, department, and state identifiers.
- Create standardized `day` and `units_sold` fields.
- Validate the expected number and uniqueness of daily sales records.

Long-format dataset created: (573900, 8)

Long-format dataset standardized.

✅ Long-format sales dataset validated successfully.

Total rows : 573,900

Unique SKU-store series: 300

Historical day range : 1 to 1913

	id	item_id	dept_id	cat_id	store_id	state_id	day	units
0	HOBBIES_1_001_CA_1_validation	HOBBIES_1_001	HOBBIES_1	HOBBIES	CA_1	CA	1	
1	HOBBIES_1_002_CA_1_validation	HOBBIES_1_002	HOBBIES_1	HOBBIES	CA_1	CA	1	
2	HOBBIES_1_003_CA_1_validation	HOBBIES_1_003	HOBBIES_1	HOBBIES	CA_1	CA	1	
3	HOBBIES_1_004_CA_1_validation	HOBBIES_1_004	HOBBIES_1	HOBBIES	CA_1	CA	1	
4	HOBBIES_1_005_CA_1_validation	HOBBIES_1_005	HOBBIES_1	HOBBIES	CA_1	CA	1	

Dataset shape : (573900, 8)

Total units : 573,932

✅ Validation Result

The development sales dataset was successfully reshaped from wide format to long format.

Validation confirmed that:

- The expected 573,900 daily sales records were generated.
- Each SKU-store series contains one unique record for every historical sales day.
- The historical day values span from 1 to 1,913.
- No missing values were introduced during the transformation.
- No negative sales quantities were detected.

The validated long-format dataset is now ready for calendar integration and subsequent feature engineering.

🔖 11.1.5 Standardize Sales Identifiers

The reshaped sales dataset currently contains multiple product and store identifier fields inherited from the original M5 dataset. To establish a consistent identifier structure across the data engineering pipeline, these fields will be standardized before further transformations.

A unique `sku_id` will be assigned to each item–store combination. This identifier will become the primary product key used throughout the project, ensuring consistent identification of every SKU across the final engineered datasets.

This step will:

- create a unique `sku_id` for each item–store combination;
- preserve the original product and store identifiers for traceability;
- verify that each sales series maps to exactly one `sku_id` ;

- validate the completeness and uniqueness of the identifier mapping.

The standardized identifier structure will provide a reliable foundation for the remaining data engineering steps.

Standardized SKU identifiers created successfully.

✓ Standardized sales identifiers validated successfully.

Unique SKU identifiers : 300

Missing SKU identifiers: 0

Duplicate mappings : 0

✓ Validation Result

The sales identifiers were successfully standardized by creating a unique `sku_id` for every item-store combination.

Validation confirmed that:

- Every development SKU-store series was assigned a unique `sku_id`.
- No missing SKU identifiers were found.
- No duplicate identifier mappings were detected.
- Each original sales series maps to exactly one standardized `sku_id`.

The validated `sku_id` is now ready to serve as the common product key across the Sales Daily, SKU Master, and Inventory Snapshot datasets.

11.1.6 Attach Calendar Keys

The standardized sales dataset currently records each observation using the historical day identifier. This step maps each sales record to its corresponding calendar date using the calendar dataset.

This step will:

- prepare a calendar lookup using the historical day identifier;
- attach the corresponding calendar date to each sales record;
- preserve all existing sales observations and identifiers;
- validate that every sales record receives a valid calendar date.

The resulting dataset will include a standardized date reference for subsequent engineering steps.

Calendar dates attached successfully.

Dataset shape: (573900, 10)

✓ Calendar date mapping validated successfully.

Total sales records : 573,900

Missing dates : 0

Duplicate day keys : 0

✓ Validation Result

Calendar dates were successfully attached to all sales records.

Validation confirmed that:

- All 573,900 sales records were mapped to valid calendar dates.
- No missing calendar dates were introduced.
- No duplicate day keys were found in the calendar lookup.
- Every historical day maps to exactly one calendar date.

The validated sales dataset now contains standardized calendar dates and is ready for weekly price integration.

11.1.7 Attach Weekly Selling Prices

Selling prices are recorded weekly for each item-store combination. This step links every daily sales record to its corresponding retail week and selling price.

This step will:

- attach the `wm_yr_wk` calendar key;
- merge prices using `item_id`, `store_id`, and `wm_yr_wk`;
- preserve all sales observations;
- validate price coverage and merge integrity.

The resulting dataset will support revenue and promotion feature engineering.

Retail week keys attached successfully.

Dataset shape: (573900, 11)

Missing week keys: 0

Weekly selling prices attached successfully.

Dataset shape: (573900, 12)

Missing prices: 81,823

Missing price diagnosis completed.

Missing price records : 81,823

Missing prices with sales : 0

Missing prices without sales: 81,823

Affected SKUs : 180

Missing selling prices handled successfully.

Remaining missing prices: 0

 Weekly selling prices validated successfully.

Total sales records : 573,900

Missing prices : 0

Validation Result

Weekly selling prices were successfully attached to all sales records.

Validation confirmed that:

- Retail week keys were mapped successfully before price integration.
- All missing selling prices occurred only during zero-sales periods.
- Missing selling prices were safely replaced with **0.0** without affecting observed sales.

- The final selling price field contains no missing values.

The validated sales dataset is now ready for revenue and promotion feature engineering.

11.1.8 Engineer Revenue and Promotion Fields

Revenue and promotional indicators provide valuable business context beyond sales volume. This step derives additional features required for sales analysis and forecasting.

This step will:

- calculate revenue for each sales record;
- create a promotion indicator when the current weekly selling price is lower than the previous available weekly price;
- preserve all existing sales information;
- validate the engineered business features.

The resulting dataset will contain the core measures required for the final sales dataset.

Revenue and promotion features engineered successfully.

Dataset shape: (573900, 14)

Promotional daily records: 2,102

Promotional SKU-weeks: 301

✅ Revenue and promotion features validated successfully.

Missing revenue values : 0

Missing promo flags : 0

Negative revenue rows : 0

Promotional daily rows : 2,102

✅ Validation Result

Revenue and promotion features were engineered successfully for all sales records.

Validation confirmed that:

- Revenue values were calculated successfully for every record.
- No missing or negative revenue values were detected.
- All promotion indicators contain valid binary values.
- A total of 2,102 daily records were identified as promotional observations.

The validated business features are now ready for the preparation of the final Sales Daily dataset.

11.1.9 Validate and Export `sales_daily_final.csv`

The final sales dataset will be prepared using the required schema and validated before export.

This step will:

- select and rename the required columns;
- verify row count, schema and data completeness;
- confirm key business measures;

- export `sales_daily_final.csv` .

The exported dataset will serve as the primary daily sales table for downstream analysis and forecasting.

Final daily sales dataset prepared successfully.

Dataset shape: (573900, 6)

Columns: ['date', 'sku_id', 'units_sold', 'revenue', 'unit_price', 'promo_flag']

✅ Final daily sales dataset validated and exported successfully.

Dataset shape : (573900, 6)

Missing values : 0

Duplicate records : 0

Dataset exported to: C:\Users\hp\Project FORESIGHT\data\processed\sales_daily_final.csv

✅ Validation Result

The final Sales Daily dataset was validated successfully and exported as `sales_daily_final.csv` .

Validation confirmed that:

- The dataset conforms to the required schema.
- No missing values remain in the final dataset.
- No duplicate `date` – `sku_id` records were found.
- All business rules were satisfied successfully.
- The export file was created successfully.

The validated Sales Daily dataset is now ready for downstream analysis, dashboard development, and demand forecasting.

11.2 Calendar Dataset Engineering

The validated Calendar dataset is transformed into a clean, standardized calendar dimension that supports temporal analysis, feature engineering, and integration with the Sales Daily dataset.

In this section, we will:

- standardize calendar attributes;
- engineer time-based analytical features;
- validate the engineered calendar dimension;
- prepare the final Calendar dataset for downstream integration.

The resulting Calendar dataset will serve as the centralized time dimension for the remaining Project FORESIGHT datasets.

11.2.1 Create Calendar Working Copy

A working copy of the raw calendar dataset is created to preserve the original data during preprocessing and feature engineering. All transformations in this section will be performed on this copy.

Objectives

- Preserve the raw dataset.
- Create an independent working DataFrame.
- Verify the dataset dimensions before processing.

Calendar working copy created successfully.
Dataset shape: (1969, 14)

✓ Validation Result

The Calendar working copy was created successfully.

Validation confirmed that:

- The working dataset preserves the original dataset dimensions.
- An independent working copy was created for all subsequent engineering tasks.
- The original Calendar dataset remains unchanged throughout the engineering process.

The validated working dataset is now ready for calendar structure inspection and feature engineering.

11.2.2 Inspect Calendar Structure

Before engineering new features, the structure of the calendar dataset is reviewed to understand its columns, data types, and overall quality. This helps identify the information available for feature engineering while confirming that the dataset has been loaded correctly.

Objectives

- Review the dataset structure.
- Examine data types and missing values.
- Identify available calendar, event, and SNAP fields.

Dataset Shape:

(1969, 14)

Column Names:

```
[ 'date',
  'wm_yr_wk',
  'weekday',
  'wday',
  'month',
  'year',
  'd',
  'event_name_1',
  'event_type_1',
  'event_name_2',
  'event_type_2',
  'snap_CA',
  'snap_TX',
  'snap_WI' ]
```

Dataset Information:

```
<class 'pandas.DataFrame'>
```

RangeIndex: 1969 entries, 0 to 1968

Data columns (total 14 columns):

#	Column	Non-Null Count	Dtype
0	date	1969 non-null	str
1	wm_yr_wk	1969 non-null	int64
2	weekday	1969 non-null	str
3	wday	1969 non-null	int64
4	month	1969 non-null	int64
5	year	1969 non-null	int64
6	d	1969 non-null	str
7	event_name_1	162 non-null	str
8	event_type_1	162 non-null	str
9	event_name_2	5 non-null	str
10	event_type_2	5 non-null	str
11	snap_CA	1969 non-null	int64
12	snap_TX	1969 non-null	int64
13	snap_WI	1969 non-null	int64

dtypes: int64(7), str(7)

memory usage: 263.1 KB

First Five Records:

	date	wm_yr_wk	weekday	wday	month	year	d	event_name_1	event_type_1	event_name_2	event_type_2	snap_CA	snap_TX	snap_WI
0	2011-01-29	11101	Saturday	1	1	2011	d_1	NaN	NaN					
1	2011-01-30	11101	Sunday	2	1	2011	d_2	NaN	NaN					
2	2011-01-31	11101	Monday	3	1	2011	d_3	NaN	NaN					
3	2011-02-01	11101	Tuesday	4	2	2011	d_4	NaN	NaN					
4	2011-02-02	11101	Wednesday	5	2	2011	d_5	NaN	NaN					

Structure Summary

The Calendar dataset structure was reviewed successfully.

The inspection confirms that:

- The dataset contains the expected calendar attributes, event information, and SNAP indicators.
- The available fields provide the required inputs for calendar standardization and feature engineering.
- No structural issues were identified that would prevent the subsequent engineering steps.

11.2.3 Standardize Calendar Fields

The existing calendar fields are standardized before creating new features. The date column is converted to datetime format, records are arranged chronologically, and the index is reset.

Objectives

- Convert the date column to datetime.
- Sort records chronologically.
- Verify date uniqueness and continuity.

Calendar fields standardized successfully.

Date data type : datetime64[us]

Date range : 2011-01-29 to 2016-06-19

Duplicate dates: 0

Missing dates : 0

✅ Validation Result

The Calendar fields were standardized successfully.

Validation confirmed that:

- All required calendar fields were renamed consistently.
- No calendar information was lost during the standardization process.
- The dataset structure and record count were preserved.
- The standardized schema is consistent with the Project FORESIGHT naming convention.

The validated Calendar dataset is now ready for calendar feature engineering.

🔧 11.2.4 Engineer Time Features

Time-based features are derived from the standardized date column to support trend analysis, seasonality detection, forecasting, and dashboard reporting.

Objectives

- Create business-friendly calendar attributes.
- Improve time-series analysis and forecasting.
- Enhance dashboard readability and reporting.

Time features engineered successfully.

New features created: 8

	date	quarter	month_name	week_of_year	day	day_name	day_of_week	is_weekend	season
0	2011-01-29	1	January	4	29	Saturday	5	1	Winter
1	2011-01-30	1	January	4	30	Sunday	6	1	Winter
2	2011-01-31	1	January	5	31	Monday	0	0	Winter
3	2011-02-01	1	February	5	1	Tuesday	1	0	Winter
4	2011-02-02	1	February	5	2	Wednesday	2	0	Winter

✅ Time features validated successfully.

```
Missing values           : 0
Invalid quarter values   : 0
Invalid month names      : 0
Invalid week values      : 0
Invalid day values       : 0
Invalid day names        : 0
Invalid day-of-week values : 0
Invalid weekend flags     : 0
Invalid season values    : 0
```

✅ Validation Result

Eight time-based features were engineered successfully from the standardized `date` column.

Validation confirmed that:

- No missing values were introduced in the engineered time features.
- Quarter, month, week, day, and weekday attributes correctly match their corresponding calendar dates.
- All weekend indicators follow the expected Saturday–Sunday rule.
- Every calendar date was assigned to the correct season.
- All engineered attributes satisfy the expected calendar business rules.

The validated time features are now ready for event, SNAP, holiday, and downstream forecasting feature engineering.

🔧 11.2.5 Engineer Event and SNAP Features

Event and SNAP features are engineered to capture calendar-based factors that may influence product demand and customer purchasing behaviour.

The `event_flag` identifies dates containing at least one recorded calendar event, while the `snap_flag` identifies dates when SNAP benefits are active in at least one of the tracked states.

Objectives

- Create an event indicator for special calendar events.
- Create a unified calendar-level SNAP indicator across California, Texas, and Wisconsin.
- Support calendar analysis, dashboard reporting, and subsequent demand-forecasting workflows.

Event and SNAP features engineered successfully.
New features created: 2

	date	event_name_1	event_name_2	event_flag	snap_CA	snap_TX	snap_WI	snap_flag
3	2011-02-01	NaN	NaN	0	1	1	0	1
4	2011-02-02	NaN	NaN	0	1	0	1	1
5	2011-02-03	NaN	NaN	0	1	1	1	1
6	2011-02-04	NaN	NaN	0	1	0	0	1
7	2011-02-05	NaN	NaN	0	1	1	1	1
8	2011-02-06	SuperBowl	NaN	1	1	1	1	1
9	2011-02-07	NaN	NaN	0	1	1	0	1
10	2011-02-08	NaN	NaN	0	1	0	1	1
11	2011-02-09	NaN	NaN	0	1	1	1	1
12	2011-02-10	NaN	NaN	0	1	0	0	1

✅ Event and SNAP features validated successfully.

Missing engineered values : 0
Invalid SNAP source values: 0
Invalid event flags : 0
Invalid SNAP flags : 0
Event logic errors : 0
SNAP logic errors : 0

✅ Validation Result

The event and unified SNAP indicators were engineered successfully.

Validation confirmed that:

- No missing values were introduced in the engineered features.
- The source SNAP fields contain valid binary values.
- Both engineered indicators contain only valid binary values.
- The `event_flag` correctly identifies dates containing at least one recorded event.

- The `snap_flag` correctly identifies dates when SNAP benefits are active in at least one tracked state.
- No event or SNAP business-rule violations were detected.

The validated event and SNAP features are now ready for final Calendar dataset preparation.

11.2.6 Prepare Final Dataset

The engineered Calendar dataset is prepared by selecting the required columns and arranging them in a consistent, business-friendly order.

Following final validation and export, this dataset will serve as the standardized calendar dimension for downstream analysis, demand forecasting, and dashboard development.

Objectives

- Select the required calendar identifiers and engineered features.
- Organize the columns into a logical and business-friendly structure.
- Prepare the Calendar dataset for final validation and export.

 Final calendar dataset prepared successfully.

Rows : 1969

Columns : 22

	date	d	wm_yr_wk	year	quarter	month	month_name	week_of_year	day	day_name	d
0	2011-01-29	d_1	11101	2011	1	1	January	4	29	Saturday	
1	2011-01-30	d_2	11101	2011	1	1	January	4	30	Sunday	
2	2011-01-31	d_3	11101	2011	1	1	January	5	31	Monday	
3	2011-02-01	d_4	11101	2011	1	2	February	5	1	Tuesday	
4	2011-02-02	d_5	11101	2011	1	2	February	5	2	Wednesday	
5	2011-02-03	d_6	11101	2011	1	2	February	5	3	Thursday	
6	2011-02-04	d_7	11101	2011	1	2	February	5	4	Friday	
7	2011-02-05	d_8	11102	2011	1	2	February	5	5	Saturday	
8	2011-02-06	d_9	11102	2011	1	2	February	5	6	Sunday	
9	2011-02-07	d_10	11102	2011	1	2	February	6	7	Monday	

Preparation Summary

The final Calendar dataset was assembled successfully with 1,969 records and 22 selected columns.

The prepared structure includes:

- calendar and retail-week identifiers;
- standardized time attributes;
- engineered season and weekend features;
- event details and the event indicator;
- state-level and unified SNAP indicators.

The prepared dataset will now undergo final schema, quality, uniqueness, data-type, and business-rule validation.

11.2.7 Validate Final Dataset

The final Calendar dataset is validated to confirm that its schema, completeness, key integrity, date continuity, data type, and business rules meet the Project FORESIGHT standards before export.

Validation Checks

- Verify record retention, dataset structure, and column order.
- Validate missing values and duplicate records.
- Confirm the uniqueness of the `date` primary key and `d` calendar identifier.
- Verify chronological order and uninterrupted daily date coverage.
- Validate all binary indicator columns.
- Confirm the standardized `date` data type.

Final Calendar validation rules defined successfully.

Expected rows : 1969

Expected columns: 22

Calendar quality and integrity metrics calculated successfully.

Required missing : 0

Duplicate rows : 0

Duplicate dates : 0

Duplicate calendar IDs : 0

Missing calendar dates : 0

 Final Calendar quality and business rules validated successfully.

✅ Final Calendar dataset validated successfully.

```
Rows : 1,969
Columns : 22
Column structure : Passed
Required missing : 0
Duplicate rows : 0
Duplicate dates : 0
Duplicate calendar IDs : 0
Chronological order : Passed
Missing calendar dates : 0
Invalid binary values : 0
Date data type : datetime64[us]
```

Binary Indicator Validation:

```
is_weekend : 0
event_flag : 0
snap_CA : 0
snap_TX : 0
snap_WI : 0
snap_flag : 0
```

✅ Validation Result

The final Calendar dataset passed all required structural, completeness, uniqueness, continuity, data-type, and business-rule validation checks.

Validation confirmed that:

- All source Calendar records were retained in the final dataset.
- The final dataset conforms to the required 22-column schema and column order.
- No missing values were found in the required fields.
- No duplicate rows, dates, or calendar identifiers were detected.
- Calendar dates are arranged chronologically with uninterrupted daily coverage.
- All weekend, event, and SNAP indicators contain valid binary values.
- The `date` field retains the required datetime data type.

The validated Calendar dataset is ready for export as `calendar_final.csv`.

11.2.8 Export `calendar_final.csv`

The Calendar dataset validated in Section 11.2.7 is exported to the project's processed data directory for use in downstream analytical and forecasting workflows.

This step will:

- confirm that the completed final validation checks remain satisfied;
- create the processed data directory when required;
- export the validated dataset as `calendar_final.csv` ;
- reload the exported file and verify its structure and content integrity;
- establish the exported file as the official Calendar dimension for Project FORESIGHT.

The verified export will support downstream feature engineering, demand forecasting, dashboard development, and SKU-level analytical workflows.

✅ Calendar dataset is ready for export.

Pre-export checks passed: 9

Calendar dataset export completed.

Export path: C:\Users\hp\Project FORESIGHT\data\processed\calendar_final.csv

✅ Exported Calendar dataset verified successfully.

✅ Final Calendar dataset exported and verified successfully.

Dataset shape : (1969, 22)

Schema preserved : Passed

Content integrity : Passed

Export file size : 156.11 KB

Dataset exported to : C:\Users\hp\Project FORESIGHT\data\processed\calendar_final.csv

✅ Validation Result

The validated Calendar dataset was exported and verified successfully as `calendar_final.csv` .

Validation confirmed that:

- All final pre-export validation requirements remained satisfied.
- The export file was created successfully in the processed data directory.
- The exported dataset preserves the required 1,969-row and 22-column structure.
- The complete schema and column order were retained.
- No unintended index column was introduced during export.
- The reloaded export contains the same values as the validated in-memory dataset.
- The exported file is complete, readable, and ready for downstream use.

The verified `calendar_final.csv` file now serves as the official Calendar dimension for subsequent Project FORESIGHT workflows.

11.3 SKU Master Dataset Engineering

The SKU Master dataset will be created by integrating product hierarchy, store identifiers, historical selling prices, and sales activity into a single SKU-store reference table.

In Project FORESIGHT, each `sku_id` represents one unique item-store combination, consistent with the standardized identifier structure established in the Sales Daily pipeline.

The engineering process includes:

- creating protected working copies of the required upstream datasets;
- constructing the standardized SKU-store reference foundation;
- deriving product lifecycle and pricing attributes;
- estimating cost and margin metrics unavailable in the source data;
- classifying SKU status using defined business rules;
- validating and exporting the final SKU Master dataset.

The completed dataset will serve as the primary SKU-store dimension for demand forecasting, inventory planning, dashboard reporting, and downstream analytical workflows in Project FORESIGHT.

11.3.1 Create Working Copies

The SKU Master pipeline uses three upstream datasets:

- the raw Sales dataset containing product, store, and historical sales information;
- the validated Calendar dataset containing date and retail-week references;
- the raw Prices dataset containing weekly item-store selling prices.

Separate working copies are created to protect these upstream DataFrames during SKU-level engineering.

This step will:

- create independent working copies of the three required datasets;
- preserve their existing dimensions and source structures;
- verify that each working DataFrame is separate from its upstream DataFrame;
- prepare the working datasets for source inspection and SKU Master construction.

The validated working copies will support all subsequent SKU Master engineering steps.

SKU Master working copies created successfully.

 SKU Master working copies validated successfully.

Working Dataset Dimensions

```
-----  
Sales dataset shape      : (30490, 1919)  
Calendar dataset shape   : (1969, 22)  
Prices dataset shape     : (6841121, 4)
```

Key Sales Entities

```
-----  
Unique raw sales series  : 30,490  
Unique items             : 3,049  
Unique stores            : 10
```

Working-Copy Independence

```
-----  
Sales      : True  
Calendar   : True  
Prices     : True
```

Validation Result

Three independent working datasets were created successfully for SKU Master engineering.

Validation confirmed that:

- Separate working copies were created for the Sales, Calendar, and Prices datasets.
- Each working DataFrame is independent of its corresponding upstream DataFrame.
- All source rows and columns were preserved during the copy operation.
- The Sales working dataset contains 30,490 raw item-store sales series, 3,049 unique items, and 10 stores.
- The upstream datasets remain protected from subsequent SKU Master transformations.

The validated working copies are now ready for source-structure inspection.

11.3.2 Inspect Source Structures

The Sales, Calendar, and Prices working datasets are inspected to confirm that the fields required for SKU Master engineering are available, complete, and structurally suitable for downstream transformations.

This step will:

- review the dimensions and required fields of each source dataset;
- verify that all required source columns are present;
- inspect representative records from the key business fields;
- confirm completeness of the required fields;
- validate the uniqueness of the source identifiers and merge keys.

The validated source structures will provide a reliable foundation for constructing the SKU Master dataset.

 All required SKU Master source fields are present.

Source Dataset Dimensions

Sales dataset shape : (30490, 1919)
Calendar dataset shape : (1969, 22)
Prices dataset shape : (6841121, 4)

Sales Reference Fields

	id	item_id	dept_id	cat_id	store_id	state_id
0	HOBBIES_1_001_CA_1_validation	HOBBIES_1_001	HOBBIES_1	HOBBIES	CA_1	CA
1	HOBBIES_1_002_CA_1_validation	HOBBIES_1_002	HOBBIES_1	HOBBIES	CA_1	CA
2	HOBBIES_1_003_CA_1_validation	HOBBIES_1_003	HOBBIES_1	HOBBIES	CA_1	CA
3	HOBBIES_1_004_CA_1_validation	HOBBIES_1_004	HOBBIES_1	HOBBIES	CA_1	CA
4	HOBBIES_1_005_CA_1_validation	HOBBIES_1_005	HOBBIES_1	HOBBIES	CA_1	CA

Calendar Reference Fields

	date	d	wm_yr_wk
0	2011-01-29	d_1	11101
1	2011-01-30	d_2	11101
2	2011-01-31	d_3	11101
3	2011-02-01	d_4	11101
4	2011-02-02	d_5	11101

Prices Reference Fields

	store_id	item_id	wm_yr_wk	sell_price
0	CA_1	HOBBIES_1_001	11325	9.58
1	CA_1	HOBBIES_1_001	11326	9.58
2	CA_1	HOBBIES_1_001	11327	8.26
3	CA_1	HOBBIES_1_001	11328	8.26
4	CA_1	HOBBIES_1_001	11329	8.26

✔ SKU Master source structures validated successfully.

Missing Sales values : 0
Missing Calendar values : 0
Missing Prices values : 0
Duplicate Sales IDs : 0
Duplicate Calendar dates : 0
Duplicate Calendar day IDs : 0
Duplicate price keys : 0

✔ Validation Result

The Sales, Calendar, and Prices working datasets passed all source-readiness checks successfully.

Validation confirmed that:

- All fields required for SKU Master engineering are present.
- No missing values were detected in the required source fields.
- Every raw Sales-series identifier is unique.
- Calendar dates and historical day identifiers are unique.
- Each item-store-week combination contains no duplicate pricing record.
- The source structures support the required SKU-level joins and feature-engineering operations.

The validated source datasets are now ready for construction of the SKU Master foundation.

11.3.3 Build SKU Master Foundation

The SKU Master foundation is created by extracting the core product, store, and hierarchy fields from the Sales working dataset. The resulting table will contain one record for every unique item-store combination.

The standardized `sku_id` will follow the canonical Project FORESIGHT identifier established in the Sales Daily pipeline:

```
item_id + "_" + store_id
```

This step will:

- extract the required product, department, store, and state reference fields;
- create the standardized `sku_id` for every item-store combination;
- standardize source column names using the project naming convention;
- arrange the foundation fields in a logical business order;
- validate identifier uniqueness, hierarchy consistency, and record completeness.

The validated foundation will support the addition of lifecycle, pricing, cost, margin, and product-status attributes.

✅ SKU Master foundation created successfully.

Dataset shape: (30490, 6)

	sku_id	item_id	category	department	store_id	state_id
0	HOBBIES_1_001_CA_1	HOBBIES_1_001	HOBBIES	HOBBIES_1	CA_1	CA
1	HOBBIES_1_002_CA_1	HOBBIES_1_002	HOBBIES	HOBBIES_1	CA_1	CA
2	HOBBIES_1_003_CA_1	HOBBIES_1_003	HOBBIES	HOBBIES_1	CA_1	CA
3	HOBBIES_1_004_CA_1	HOBBIES_1_004	HOBBIES	HOBBIES_1	CA_1	CA
4	HOBBIES_1_005_CA_1	HOBBIES_1_005	HOBBIES	HOBBIES_1	CA_1	CA
5	HOBBIES_1_006_CA_1	HOBBIES_1_006	HOBBIES	HOBBIES_1	CA_1	CA
6	HOBBIES_1_007_CA_1	HOBBIES_1_007	HOBBIES	HOBBIES_1	CA_1	CA
7	HOBBIES_1_008_CA_1	HOBBIES_1_008	HOBBIES	HOBBIES_1	CA_1	CA
8	HOBBIES_1_009_CA_1	HOBBIES_1_009	HOBBIES	HOBBIES_1	CA_1	CA
9	HOBBIES_1_010_CA_1	HOBBIES_1_010	HOBBIES	HOBBIES_1	CA_1	CA

✅ SKU Master foundation validated successfully.

```
Dataset shape           : (30490, 6)
Unique canonical SKU IDs : 30,490
Missing values          : 0
Duplicate rows           : 0
Duplicate SKU IDs        : 0
Duplicate item-store pairs : 0
Invalid canonical IDs     : 0
Hierarchy mapping errors  : 0
Store-state mapping errors : 0
```

✅ Validation Result

The SKU Master foundation was created and validated successfully.

Validation confirmed that:

- One foundation record was created for every unique item-store combination.
- All `sku_id` values follow the canonical Project FORESIGHT identifier convention.
- The standardized identifiers are complete and unique.
- No source records were lost during foundation construction.
- No missing values, duplicate records, or duplicate item-store combinations were detected.
- Each item maps consistently to one category and department.
- Each store maps consistently to one state.
- No unsupported or duplicated hierarchy field was introduced.

The validated SKU Master foundation is now ready for lifecycle, pricing, cost, margin, and product-status feature engineering.

11.3.4 Engineer SKU Lifecycle Fields

SKU lifecycle attributes are derived from historical price availability and positive sales activity. These fields describe when each item-store combination became commercially observable, when it first and last recorded sales, and how long it existed within the observed sales period.

Because the source data does not contain an explicit product-launch date, `launch_date` is treated as an operational proxy defined as the first calendar date of the earliest retail week containing a positive selling price.

This step will:

- prepare a complete mapping between historical sales-day columns and calendar dates;
- estimate the launch date for each SKU from its earliest positive-price week;
- identify the first and last dates on which each SKU recorded positive sales;
- calculate SKU age at the end of the observed sales period;
- validate lifecycle completeness, chronological order, identifier integrity, and record preservation.

The validated lifecycle attributes will support SKU-status classification, inventory planning, and downstream business analysis.

✅ SKU lifecycle references prepared successfully.

Daily sales columns : 1,913
Missing date mappings : 0
First observed sales date : 2011-01-29
Last observed sales date : 2016-04-24

✅ SKU launch dates engineered successfully.

Earliest launch date : 2011-01-29
Latest launch date : 2016-02-13
Missing launch dates : 0

✅ First and last sale dates engineered successfully.

Lifecycle records created : 30,490
Missing first-sale dates : 0
Missing last-sale dates : 0
Rows preserved : 30,490

✅ SKU age calculated successfully.

Minimum SKU age : 71 days
Maximum SKU age : 1,912 days

	sku_id	store_id	launch_date	first_sale_date	last_sale_date	sku_age_days
0	HOBBIES_1_001_CA_1	CA_1	2013-07-13	2013-07-18	2016-04-24	1016
1	HOBBIES_1_002_CA_1	CA_1	2011-06-18	2011-06-21	2016-04-20	1772
2	HOBBIES_1_003_CA_1	CA_1	2014-02-01	2014-02-07	2016-04-24	813
3	HOBBIES_1_004_CA_1	CA_1	2011-03-05	2011-03-06	2016-04-24	1877
4	HOBBIES_1_005_CA_1	CA_1	2011-05-21	2011-05-21	2016-04-24	1800
5	HOBBIES_1_006_CA_1	CA_1	2012-03-31	2012-04-02	2016-04-22	1485
6	HOBBIES_1_007_CA_1	CA_1	2012-07-07	2012-07-12	2016-04-24	1387
7	HOBBIES_1_008_CA_1	CA_1	2011-01-29	2011-01-29	2016-04-24	1912
8	HOBBIES_1_009_CA_1	CA_1	2011-01-29	2011-01-29	2016-04-19	1912
9	HOBBIES_1_010_CA_1	CA_1	2011-01-29	2011-01-31	2016-04-24	1912

✓ Validation Result

The SKU lifecycle fields were engineered and validated successfully for all item-store combinations.

Validation confirmed that:

- All historical sales-day columns were mapped to complete and chronological Calendar dates.
- Estimated launch dates were derived from the earliest retail weeks containing positive selling prices.
- First and last sale dates were identified from positive historical sales activity.
- All lifecycle records use the canonical Project FORESIGHT `sku_id`.
- No missing lifecycle values or duplicate SKU identifiers were detected.
- No sales occurred before the estimated launch date or outside the observed sales period.
- All first-sale and last-sale dates follow the required chronological sequence.
- SKU-age values are complete, non-negative, and correctly calculated.
- All SKU Master foundation records were preserved during lifecycle engineering.

The validated SKU lifecycle attributes are now ready for pricing feature engineering.

🔧 11.3.5 Engineer Pricing Fields

SKU-level pricing attributes are derived from the historical weekly selling prices available for each item-store combination.

The `avg_price` field represents the unweighted average of the available weekly selling-price records, while `max_observed_price` represents the highest selling price recorded during the available pricing history. It should be interpreted as a historical reference-price proxy rather than a confirmed official list price.

This step will:

- validate the completeness and positivity of the source selling-price records;
- calculate the average historical weekly selling price for each SKU;
- determine the maximum observed selling price for each SKU;
- create canonical SKU identifiers for the aggregated pricing records;
- integrate the pricing attributes into the SKU Master dataset;
- validate pricing coverage, logical consistency, and merge integrity.

The validated pricing attributes will support profitability analysis, cost estimation, and downstream business reporting.

✓ SKU pricing metrics calculated successfully.

Pricing SKU records : 30,490
Missing source prices : 0
Non-positive source prices: 0
Duplicate pricing SKU IDs : 0

✅ Pricing attributes integrated successfully.

SKU Master shape : (30490, 12)
SKUs updated : 30,490
Unmatched Master SKUs : 0
Unused pricing SKUs : 0
Row count preserved : True

	sku_id	avg_price	max_observed_price
0	HOBBIES_1_001_CA_1	8.29	9.58
1	HOBBIES_1_002_CA_1	3.97	3.97
2	HOBBIES_1_003_CA_1	2.97	2.97
3	HOBBIES_1_004_CA_1	4.53	4.64
4	HOBBIES_1_005_CA_1	2.94	3.08
5	HOBBIES_1_006_CA_1	0.99	1.00
6	HOBBIES_1_007_CA_1	7.89	7.94
7	HOBBIES_1_008_CA_1	0.48	0.50
8	HOBBIES_1_009_CA_1	1.76	1.77
9	HOBBIES_1_010_CA_1	2.98	3.17

✅ SKU pricing fields validated successfully.

Missing pricing values : 0
Invalid price relationships : 0
Non-positive engineered prices : 0
Numeric pricing data types : True
Duplicate Master SKU IDs : 0
Pricing row count preserved : True

✅ Validation Result

The SKU pricing attributes were engineered and validated successfully for all item-store combinations.

Validation confirmed that:

- All source weekly selling-price records were complete and positive.
- One aggregated pricing record was created for every canonical SKU.
- `avg_price` represents the unweighted average of the available weekly selling prices.
- `max_observed_price` represents the highest selling price recorded for each SKU.
- Every SKU Master record received complete pricing attributes.
- No unmatched or unused pricing SKU records were detected.
- Average prices do not exceed their corresponding maximum observed prices.
- All engineered pricing values are numeric and positive.
- Canonical SKU identifiers remained unique.
- The SKU Master row count was preserved during pricing integration.

The validated pricing attributes are now ready for cost and gross-margin estimation.

💰 11.3.6 Estimate Cost and Margin Fields

Estimated cost and margin attributes are derived from the engineered `max_observed_price` using a documented business assumption.

Because actual procurement-cost data is unavailable, the maximum observed selling price is used as a reference-price proxy. The estimated unit cost is assumed to represent 70% of this reference price, while the remaining amount represents the estimated gross margin per unit.

These fields are analytical estimates and should not be interpreted as actual procurement costs or realized accounting margins.

This step will:

- define and validate the cost-estimation assumption;
- estimate the unit cost for every canonical SKU;
- calculate the corresponding estimated gross margin per unit;
- validate the calculations, financial relationships, and record preservation.

The validated estimated financial attributes will support profitability-oriented analysis, inventory valuation assumptions, and downstream business reporting.

✅ Estimated cost and margin fields engineered successfully.

Cost estimation rate : 70%
Dataset shape : (30490, 14)
SKUs updated : 30,490
Missing reference prices : 0

	sku_id	max_observed_price	estimated_unit_cost	estimated_gross_margin
0	HOBBIES_1_001_CA_1	9.58	6.71	2.87
1	HOBBIES_1_002_CA_1	3.97	2.78	1.19
2	HOBBIES_1_003_CA_1	2.97	2.08	0.89
3	HOBBIES_1_004_CA_1	4.64	3.25	1.39
4	HOBBIES_1_005_CA_1	3.08	2.16	0.92
5	HOBBIES_1_006_CA_1	1.00	0.70	0.30
6	HOBBIES_1_007_CA_1	7.94	5.56	2.38
7	HOBBIES_1_008_CA_1	0.50	0.35	0.15
8	HOBBIES_1_009_CA_1	1.77	1.24	0.53
9	HOBBIES_1_010_CA_1	3.17	2.22	0.95

✅ Estimated cost and margin fields validated successfully.

Missing financial values : 0
Non-positive estimated costs : 0
Non-positive estimated margins : 0
Invalid cost relationships : 0
Invalid cost calculations : 0
Invalid margin calculations : 0
Reconciliation errors : 0
Numeric financial data types : True
Financial row count preserved : True

✅ Validation Result

The estimated SKU cost and margin fields were engineered and validated successfully.

Validation confirmed that:

- The documented 70% cost-estimation assumption was applied consistently.
- All calculations use `max_observed_price` as the reference-price proxy.
- Estimated unit costs and gross margins are complete, numeric, and positive.
- Every estimated unit cost remains below its corresponding reference price.
- Estimated unit costs were calculated accurately from the documented cost rate.
- Estimated gross margins were calculated accurately from the reference price and estimated cost.
- Estimated cost and gross margin reconcile to the maximum observed price within the required rounding tolerance.
- Canonical SKU identifiers remained unique.
- The SKU Master row count was preserved during financial feature engineering.

These fields are analytical estimates rather than actual procurement costs or realized accounting margins. The validated SKU Master is now ready for SKU-status classification.

11.3.7 Classify SKU Status

SKU status is classified using the recency of positive sales activity for each item-store combination.

The classification represents a historical snapshot as of the final date in the observed sales period. It does not represent the SKU's current real-world commercial status.

The following Project FORESIGHT business rules are applied:

- **Active:** the last recorded sale occurred within the previous 30 days;
- **Slow Moving:** the last recorded sale occurred between 31 and 90 days ago;
- **Inactive:** the last recorded sale occurred more than 90 days ago.

This step will:

- calculate the number of days since the last recorded sale for each SKU;
- classify each SKU using documented recency thresholds;
- retain the recency measure for reporting and classification traceability;
- validate classification completeness, rule accuracy, identifier uniqueness, and record preservation.

The validated SKU-status attributes will support inventory monitoring, assortment analysis, and downstream business reporting.

✅ SKU status classified successfully.

Status reference date : 2016-04-24

Active threshold : 0-30 days

Slow Moving threshold : 31-90 days

Inactive threshold : More than 90 days

SKUs classified : 30,490

	sku_id	last_sale_date	days_since_last_sale	sku_status
0	HOBBIES_1_001_CA_1	2016-04-24	0	Active
1	HOBBIES_1_002_CA_1	2016-04-20	4	Active
2	HOBBIES_1_003_CA_1	2016-04-24	0	Active
3	HOBBIES_1_004_CA_1	2016-04-24	0	Active
4	HOBBIES_1_005_CA_1	2016-04-24	0	Active
5	HOBBIES_1_006_CA_1	2016-04-22	2	Active
6	HOBBIES_1_007_CA_1	2016-04-24	0	Active
7	HOBBIES_1_008_CA_1	2016-04-24	0	Active
8	HOBBIES_1_009_CA_1	2016-04-19	5	Active
9	HOBBIES_1_010_CA_1	2016-04-24	0	Active

✅ SKU status validated successfully.

Missing sales recency : 0

Negative sales recency : 0

Missing SKU status : 0

Invalid status values : 0

Incorrect assignments : 0

Duplicate SKU IDs : 0

Row count preserved : True

SKU Status Distribution:

sku_status

Active 29015

Slow Moving 1097

Inactive 378

Name: count, dtype: int64[pyarrow]

SKU Status Percentage (%):

sku_status

Active 95.16

Slow Moving 3.60

Inactive 1.24

Name: proportion, dtype: double[pyarrow]

✅ Validation Result

The SKU-status attributes were engineered and validated successfully for all 30,490 item-store combinations.

Validation confirmed that:

- Sales recency was calculated as of 24 April 2016.
- All `days_since_last_sale` values are complete and non-negative.
- Every SKU was classified using the documented 30-day and 90-day thresholds.
- No missing, invalid, or incorrectly assigned status values were detected.
- All canonical `sku_id` values remained unique.
- The SKU Master row count was preserved during classification.
- The final status distribution contains 29,015 Active SKUs, 1,097 Slow Moving SKUs, and 378 Inactive SKUs.
- Active, Slow Moving, and Inactive SKUs represent 95.16%, 3.60%, and 1.24% of the complete SKU population, respectively.

The validated SKU Master is now ready for final dataset preparation.

11.3.8 Prepare Final SKU Master Dataset

The completed SKU Master dataset is prepared by selecting and arranging the validated reference, lifecycle, pricing, financial, and status fields in a consistent business-ready structure.

Each record represents one canonical item-store combination identified by `sku_id`. The final schema retains the engineered sales-recency field for status traceability and explicitly labels price, cost, and margin estimates according to their analytical meaning.

This step will:

- define the required final SKU Master schema;
- confirm that all required engineered fields are available;
- select and arrange the fields in a logical business order;
- sort the dataset using the canonical SKU identifier;
- create an independent final dataset for inspection and export.

The prepared dataset will serve as the primary SKU-store reference table for inventory analysis, forecasting, dashboard reporting, and downstream analytical workflows.

 Final SKU Master dataset prepared successfully.

Dataset shape : (30490, 16)
Total SKUs : 30,490
Columns selected : 16
Rows preserved : True

	sku_id	item_id	category	department	store_id	state_id	launch_date	first_sale_date
0	FOODS_1_001_CA_1	FOODS_1_001	FOODS	FOODS_1	CA_1	CA	2011-01-29	2011-01-29
1	FOODS_1_001_CA_2	FOODS_1_001	FOODS	FOODS_1	CA_2	CA	2011-01-29	2011-01-29
2	FOODS_1_001_CA_3	FOODS_1_001	FOODS	FOODS_1	CA_3	CA	2011-01-29	2011-01-29
3	FOODS_1_001_CA_4	FOODS_1_001	FOODS	FOODS_1	CA_4	CA	2011-01-29	2011-01-29
4	FOODS_1_001_TX_1	FOODS_1_001	FOODS	FOODS_1	TX_1	TX	2011-01-29	2011-01-29
5	FOODS_1_001_TX_2	FOODS_1_001	FOODS	FOODS_1	TX_2	TX	2011-01-29	2011-01-29
6	FOODS_1_001_TX_3	FOODS_1_001	FOODS	FOODS_1	TX_3	TX	2011-02-05	2011-02-05
7	FOODS_1_001_WI_1	FOODS_1_001	FOODS	FOODS_1	WI_1	WI	2011-01-29	2011-01-29
8	FOODS_1_001_WI_2	FOODS_1_001	FOODS	FOODS_1	WI_2	WI	2011-01-29	2011-01-29
9	FOODS_1_001_WI_3	FOODS_1_001	FOODS	FOODS_1	WI_3	WI	2011-01-29	2011-01-29

```
<class 'pandas.DataFrame'>
RangeIndex: 30490 entries, 0 to 30489
Data columns (total 16 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   sku_id                                30490 non-null  object
1   item_id                               30490 non-null  str
2   category                              30490 non-null  str
3   department                            30490 non-null  str
4   store_id                              30490 non-null  str
5   state_id                              30490 non-null  str
6   launch_date                           30490 non-null  datetime64[us]
7   first_sale_date                       30490 non-null  datetime64[ns]
8   last_sale_date                        30490 non-null  datetime64[ns]
9   sku_age_days                          30490 non-null  Int32
10  days_since_last_sale                  30490 non-null  Int32
11  avg_price                             30490 non-null  float64
12  max_observed_price                    30490 non-null  float64
13  estimated_unit_cost                   30490 non-null  float64
14  estimated_gross_margin                30490 non-null  float64
15  sku_status                            30490 non-null  string
dtypes: Int32(2), datetime64[ns](2), datetime64[us](1), float64(4), object(1), str(5), string(1)
memory usage: 4.7+ MB
```

Preparation Summary

The final SKU Master dataset was prepared successfully in a consistent business-ready structure.

Preparation confirmed that:

- One record was retained for every canonical item-store combination.
- All required reference, lifecycle, pricing, financial, and status fields were available.
- The unsupported duplicate `subcategory` field was excluded.
- `days_since_last_sale` was retained to provide status-classification traceability.

- Historical reference-price, cost, and margin estimates were labelled explicitly.
- The canonical `sku_id` was retained as the primary dataset key.
- All working SKU records were preserved during final preparation.
- The records were arranged in a deterministic SKU-level order.

The prepared SKU Master dataset is now ready for final quality validation and export.

11.3.9 Validate and Export SKU Master Dataset

The completed SKU Master dataset will undergo a final end-to-end quality validation before being exported to the processed data directory.

This stage confirms that the final dataset preserves the required schema, canonical identifiers, lifecycle relationships, pricing logic, estimated financial calculations, and SKU-status classifications. The exported CSV will then be reloaded and compared with the in-memory dataset to verify successful round-trip preservation.

This step will:

- validate the final schema, record count, completeness, and identifier uniqueness;
- confirm canonical SKU construction and hierarchy consistency;
- validate lifecycle, pricing, estimated financial, and status business rules;
- export the validated dataset to the processed data directory;
- reload the exported file and verify its structure and content.

The successfully verified export will serve as the official SKU-store reference table for downstream Project FORESIGHT workflows.

✅ Final SKU Master structure validated successfully.

Rows	: 30,490
Columns	: 16
Column structure	: Passed
Missing values	: 0
Duplicate rows	: 0
Duplicate SKU IDs	: 0
Duplicate item-store pairs	: 0
Invalid canonical IDs	: 0
SKU ordering	: Passed
Date data types	: Passed
Numeric data types	: Passed
Hierarchy mapping errors	: 0
Store-state mapping errors	: 0

✅ Final SKU Master business rules validated successfully.

```
Observation period      : 2011-01-29 to 2016-04-24
Invalid sale sequences   : 0
Sales before launch      : 0
Future launch dates     : 0
Negative SKU ages        : 0
Invalid SKU-age calculations : 0
Invalid recency calculations : 0
Non-positive prices      : 0
Invalid price relationships : 0
Invalid cost calculations : 0
Invalid margin calculations : 0
Financial reconciliation errors : 0
Invalid status values    : 0
Incorrect status assignments : 0
```

✅ Final SKU Master dataset exported and verified successfully.

```
Dataset shape           : (30490, 16)
Exported rows           : 30,490
Exported columns        : 16
Missing after reload    : 0
Duplicate SKU IDs       : 0
Duplicate rows          : 0
File size               : 3,839,729 bytes
Dataset exported to     : C:\Users\hp\Project FORESIGHT\data\processed\sku_master_final.csv
```

✅ Validation Result

The final SKU Master dataset passed all structural, integrity, business-rule, and export-verification checks successfully.

Validation confirmed that:

- The dataset contains one record for each of the 30,490 canonical item-store combinations.
- The required 16-column business-ready schema was preserved.
- No missing values, duplicate records, duplicate identifiers, or duplicate item-store combinations were detected.
- All `sku_id` values follow the canonical Project FORESIGHT identifier convention.
- Product hierarchy and store-state mappings are internally consistent.
- Lifecycle dates, SKU ages, and sales-recency calculations satisfy the required chronological rules.
- All historical pricing values are positive and logically consistent.
- Estimated unit costs and gross margins follow the documented 70% cost assumption.
- All estimated financial fields reconcile with the maximum observed reference price.
- Every SKU-status classification follows the documented 30-day and 90-day recency thresholds.
- The exported CSV was reloaded successfully with its complete structure and content preserved.
- No missing values or duplicate records were introduced during export.

The verified `sku_master_final.csv` file is ready to serve as the official SKU-store reference dataset for downstream Project FORESIGHT workflows.



11.4 Inventory Snapshot Dataset Engineering

The M5 source data does not contain observed inventory balances, purchase orders, stock receipts, or stockout records. Therefore, this section creates a simulated Inventory Snapshot dataset using historical demand and documented inventory-policy assumptions.

The engineering workflow will:

- define the simulation assumptions;
- select the development SKU population;
- construct daily inventory movements;
- engineer replenishment and inventory-health metrics;
- validate and export the final dataset.

All stock levels, replenishment activity, lost sales, and inventory values produced in this section are analytical estimates rather than observed retailer records.

11.4.1 Define Inventory Simulation Assumptions

This step defines the SKU scope, demand coverage, safety stock, replenishment lead time, and reorder-quantity policy used throughout the inventory simulation.

These values are documented analytical assumptions created for Project FORESIGHT and do not represent observed M5 retailer policies.

Inventory Simulation Assumptions

```
-----
Inventory SKU population      : 300
Base demand coverage         : 7 days
Safety-stock coverage        : 3 days
Replenishment lead time      : 5 days
Target coverage horizon      : 15 days
Reorder quantity multiplier   : 2.0x
```

 Inventory simulation assumptions validated successfully.

```
Inventory SKU population      : 300
Demand coverage days         : 7
Safety-stock days            : 3
Lead-time days               : 5
Target coverage days         : 15
Reorder quantity multiplier   : 2.0x
```

Validation Result

The inventory-simulation assumptions were defined and validated successfully.

Validation confirmed that:

- The simulation scope contains 300 development SKUs.
- Base demand coverage was configured at seven days.
- Safety-stock coverage was configured at three additional days.
- Replenishment lead time was configured at five days.
- The complete target-stock horizon equals 15 days of demand coverage.
- The replenishment-quantity multiplier was configured at 2.0 times the target-stock level.

- All parameters contain valid data types and satisfy the required logical constraints.

These parameters are documented analytical assumptions rather than observed retailer inventory policies. The validated configuration is ready for selection of the inventory-simulation SKU population.

11.4.2 Select Simulation SKU Population

This step matches the 300 canonical SKUs retained in Sales Daily with their validated SKU Master records.

The resulting population provides the product hierarchy and estimated cost attributes required for inventory simulation while preserving complete and unique SKU coverage.

 Inventory simulation SKU population selected successfully.

Available Sales Daily SKUs : 300

Matched SKU Master records : 300

	sku_id	item_id	category	department	store_id	state_id	estimated_unit_cost
0	FOODS_1_001_CA_1	FOODS_1_001	FOODS	FOODS_1	CA_1	CA	1.57
1	FOODS_1_001_CA_2	FOODS_1_001	FOODS	FOODS_1	CA_2	CA	1.57
2	FOODS_1_001_CA_3	FOODS_1_001	FOODS	FOODS_1	CA_3	CA	1.57
3	FOODS_1_001_CA_4	FOODS_1_001	FOODS	FOODS_1	CA_4	CA	1.57
4	FOODS_1_001_TX_1	FOODS_1_001	FOODS	FOODS_1	TX_1	TX	1.57
5	FOODS_1_001_TX_2	FOODS_1_001	FOODS	FOODS_1	TX_2	TX	1.57
6	FOODS_1_001_TX_3	FOODS_1_001	FOODS	FOODS_1	TX_3	TX	1.57
7	FOODS_1_001_WI_1	FOODS_1_001	FOODS	FOODS_1	WI_1	WI	1.57
8	FOODS_1_001_WI_2	FOODS_1_001	FOODS	FOODS_1	WI_2	WI	1.57
9	FOODS_1_001_WI_3	FOODS_1_001	FOODS	FOODS_1	WI_3	WI	1.57

 Inventory simulation SKU population validated successfully.

Expected SKUs : 300

Selected SKUs : 300

Duplicate SKUs : 0

Missing values : 0

SKUs missing from Master : 0

Missing selected SKUs : 0

Extra selected SKUs : 0

Invalid canonical IDs : 0

Non-positive cost estimates: 0

Validation Result

The inventory-simulation SKU population was selected and validated successfully.

Validation confirmed that:

- All 300 canonical SKUs retained in Sales Daily were represented.
- Every Sales Daily SKU was matched directly to its validated SKU Master record.
- No random resampling or temporary identifier reconstruction was required.
- No missing, duplicate, unmatched, or unexpected SKU records were detected.
- All selected identifiers follow the canonical Project FORESIGHT item-store convention.
- Product hierarchy, store, state, and estimated unit-cost attributes are complete.
- All estimated unit-cost values are positive.
- The selected SKU set exactly matches the complete Sales Daily development population.

The validated SKU population is ready for construction of the daily inventory working dataset.

11.4.3 Create Daily Inventory Working Dataset

The daily inventory working dataset combines historical demand for the selected SKUs with the estimated cost information required for inventory simulation.

This step will extract daily SKU demand, integrate estimated unit costs, arrange records chronologically, and validate complete SKU-date coverage.

The completed dataset will provide the input for demand-policy engineering and sequential inventory simulation.

✅ Daily inventory working dataset created successfully.

Dataset shape : (573900, 5)

Unique SKUs : 300

Date range : 2011-01-29 to 2016-04-24

✅ Daily inventory working dataset validated successfully.

Expected records : 573,900

Actual records : 573,900

Unique SKUs : 300

Unique dates : 1,913

Missing values : 0

Duplicate SKU-dates : 0

Incomplete histories : 0

Invalid selected SKUs : 0

Chronological order : Passed

✅ Validation Result

The daily inventory working dataset was created and validated successfully.

All 300 selected SKUs retained complete chronological demand histories, with no missing values, duplicate SKU-date records, unmatched identifiers, or invalid demand and cost values.

The dataset is ready for demand and replenishment parameter engineering.

11.4.4 Engineer Inventory Parameters

SKU-level inventory parameters will be estimated from historical average daily demand and the validated simulation assumptions.

This step will calculate reorder points, target stock levels, and fixed replenishment quantities for each simulated SKU.

The validated parameters will guide the sequential inventory simulation.

✅ Demand and replenishment parameters engineered successfully.
Total SKUs : 300

	sku_id	average_daily_demand	lead_time_days	reorder_point	target_stock	reorder_q
0	FOODS_1_001_CA_1	0.79	5	7	12	
1	FOODS_1_001_CA_2	1.16	5	10	18	
2	FOODS_1_001_CA_3	1.20	5	10	19	
3	FOODS_1_001_CA_4	0.36	5	3	6	
4	FOODS_1_001_TX_1	0.58	5	5	9	
5	FOODS_1_001_TX_2	0.58	5	5	9	
6	FOODS_1_001_TX_3	0.41	5	4	7	
7	FOODS_1_001_WI_1	0.57	5	5	9	
8	FOODS_1_001_WI_2	0.46	5	4	7	
9	FOODS_1_001_WI_3	0.29	5	3	5	

✅ Demand and replenishment parameters validated successfully.

Expected SKUs : 300
Actual SKUs : 300
Missing values : 0
Duplicate SKUs : 0
Incorrect reorder points : 0
Incorrect target stocks : 0
Incorrect reorder quantity : 0
Average demand range : 0.01 - 9.78
Reorder point range : 1 - 79
Target stock range : 1 - 147
Reorder quantity range : 2 - 294

✅ Validation Result

Demand and replenishment parameters were engineered and validated successfully for all 300 SKUs.

All records were complete and unique, and the reorder points, target stock levels, and replenishment quantities matched the documented simulation policy.

The parameters are ready for daily inventory movement simulation.

11.4.5 Simulate Daily Inventory Movements

Daily inventory movements will be simulated chronologically for each SKU using historical demand and the engineered replenishment parameters.

Stock receipts will be processed before demand, while new orders will be placed after evaluating the closing inventory position. Only one replenishment order may remain outstanding for a SKU at a time.

The completed simulation will provide one inventory-movement record for every SKU-date combination.

✅ Daily inventory movements simulated successfully.

Dataset shape : (573900, 16)

Unique SKUs : 300

Total demand : 573,932

Units fulfilled: 549,943

Lost sales : 23,989

	date	sku_id	opening_stock	units_sold	fulfilled_units	lost_sales_units	received_sto
0	2011-01-29	FOODS_1_001_CA_1	12	3	3	0	
1	2011-01-30	FOODS_1_001_CA_1	9	0	0	0	
2	2011-01-31	FOODS_1_001_CA_1	9	0	0	0	
3	2011-02-01	FOODS_1_001_CA_1	9	1	1	0	
4	2011-02-02	FOODS_1_001_CA_1	8	4	4	0	
5	2011-02-03	FOODS_1_001_CA_1	4	2	2	0	
6	2011-02-04	FOODS_1_001_CA_1	2	0	0	0	
7	2011-02-05	FOODS_1_001_CA_1	2	2	2	0	
8	2011-02-06	FOODS_1_001_CA_1	0	0	0	0	
9	2011-02-07	FOODS_1_001_CA_1	0	0	0	0	



✅ Daily inventory movements validated successfully.

Expected records : 573,900

Actual records : 573,900

Unique SKUs : 300

Missing values : 0

Duplicate SKU-dates : 0

Inventory balance errors : 0

Demand balance errors : 0

Continuity errors : 0

Receipt timing errors : 0

Order placement errors : 0

✅ Validation Result

Daily inventory movements were simulated and validated successfully for all 300 SKUs.

Inventory balances, demand fulfilment, stock continuity, order placement, receipt timing, and non-negativity checks all passed. The simulated movements are ready for inventory-status and business-metric engineering.

🔧 11.4.6 Engineer Inventory Metrics

The simulated inventory movements will be enriched with inventory-status, estimated inventory-value, and days-of-supply metrics.

These fields will support inventory-health monitoring and downstream reporting.

✅ Inventory status and business metrics engineered successfully.

Dataset shape : (573900, 20)

Inventory statuses : 4

	date	sku_id	closing_stock	average_daily_demand	inventory_status	estimated_inve
0	2011-01-29	FOODS_1_001_CA_1	9	0.79	Healthy	
1	2011-01-30	FOODS_1_001_CA_1	9	0.79	Healthy	
2	2011-01-31	FOODS_1_001_CA_1	9	0.79	Healthy	
3	2011-02-01	FOODS_1_001_CA_1	8	0.79	Healthy	
4	2011-02-02	FOODS_1_001_CA_1	4	0.79	Low Stock	
5	2011-02-03	FOODS_1_001_CA_1	2	0.79	Low Stock	
6	2011-02-04	FOODS_1_001_CA_1	2	0.79	Low Stock	
7	2011-02-05	FOODS_1_001_CA_1	0	0.79	Stockout	
8	2011-02-06	FOODS_1_001_CA_1	0	0.79	Stockout	
9	2011-02-07	FOODS_1_001_CA_1	24	0.79	Overstock	



✅ Inventory status and business metrics validated successfully.

```
Expected records      : 573,900
Actual records        : 573,900
Unique SKUs           : 300
Missing values        : 0
Duplicate SKU-dates    : 0
Invalid status values : 0
Incorrect classifications : 0
Inventory value errors : 0
Days-of-supply errors  : 0
```

Inventory Status Distribution:

```
inventory_status
Healthy      267673
Low Stock    67528
Overstock    223381
Stockout     15318
Name: count, dtype: int64[pyarrow]
```

✅ Validation Result

Inventory status and business metrics were engineered and validated successfully.

All records retained valid status classifications, estimated inventory values, and days-of-supply calculations, with no missing values, duplicate SKU-date records, or calculation errors.

The completed simulation dataset is ready for final preparation and export.



11.4.7 Prepare Final Inventory Dataset

The completed inventory simulation dataset will be arranged into its final analytical structure.

This step will retain the required movement, replenishment, inventory-status, and business-metric fields before final validation and export.

✅ Final inventory snapshot dataset prepared successfully.

```
Dataset shape : (573900, 17)
Unique SKUs   : 300
Date range    : 2011-01-29 to 2016-04-24
Rows preserved: True
```

	date	sku_id	opening_stock	units_sold	fulfilled_units	lost_sales_units	received_sto
0	2011-01-29	FOODS_1_001_CA_1	12	3	3	0	
1	2011-01-30	FOODS_1_001_CA_1	9	0	0	0	
2	2011-01-31	FOODS_1_001_CA_1	9	0	0	0	
3	2011-02-01	FOODS_1_001_CA_1	9	1	1	0	
4	2011-02-02	FOODS_1_001_CA_1	8	4	4	0	
5	2011-02-03	FOODS_1_001_CA_1	4	2	2	0	
6	2011-02-04	FOODS_1_001_CA_1	2	0	0	0	
7	2011-02-05	FOODS_1_001_CA_1	2	2	2	0	
8	2011-02-06	FOODS_1_001_CA_1	0	0	0	0	
9	2011-02-07	FOODS_1_001_CA_1	0	0	0	0	

Preparation Summary

The final Inventory Snapshot dataset was prepared successfully.

All 573,900 simulated SKU-date records were retained in a consistent 17-column structure. The dataset is ready for final validation and export.

11.4.8 Validate and Export Inventory Snapshots

The final Inventory Snapshot dataset will undergo structural and business-rule validation before export.

The exported CSV will then be reloaded and compared with the validated in-memory dataset to confirm successful preservation.

✅ Final inventory snapshot dataset validated successfully.

Rows : 573,900
Columns : 17
Unique SKUs : 300
Unique dates : 1,913
Missing values : 0
Duplicate SKU-dates : 0
Inventory balance errors : 0
Demand balance errors : 0
Invalid status values : 0
Invalid order quantities : 0
Negative analytical values : 0

✅ Inventory snapshot dataset exported and verified successfully.

Export file : inventory_snapshots_final.csv
Exported rows : 573,900
Exported columns : 17
Unique SKUs : 300
Missing values : 0
Duplicate SKU-dates : 0
Inventory errors : 0
Demand errors : 0
File size : 44,351,238 bytes
Export location : C:\Users\hp\Project FORESIGHT\data\processed\inventory_snapshots_f
inal.csv

	date	sku_id	opening_stock	units_sold	fulfilled_units	lost_sales_units	received_sto
0	2011-01-29	FOODS_1_001_CA_1	12	3	3	0	
1	2011-01-30	FOODS_1_001_CA_1	9	0	0	0	
2	2011-01-31	FOODS_1_001_CA_1	9	0	0	0	
3	2011-02-01	FOODS_1_001_CA_1	9	1	1	0	
4	2011-02-02	FOODS_1_001_CA_1	8	4	4	0	
5	2011-02-03	FOODS_1_001_CA_1	4	2	2	0	
6	2011-02-04	FOODS_1_001_CA_1	2	0	0	0	
7	2011-02-05	FOODS_1_001_CA_1	2	2	2	0	
8	2011-02-06	FOODS_1_001_CA_1	0	0	0	0	
9	2011-02-07	FOODS_1_001_CA_1	0	0	0	0	



✅ Validation Result

The Inventory Snapshot dataset passed all final validation and export-verification checks.

All simulated SKU-date records, inventory balances, demand balances, status values, and replenishment fields were preserved successfully in `inventory_snapshots_final.csv`.

The exported dataset is ready for downstream Project FORESIGHT workflows.



12. Final Dataset Validation

This section performs notebook-level validation across all engineered and exported datasets.

It verifies dataset availability, schema integrity, key uniqueness, cross-dataset consistency, and readiness for downstream Project FORESIGHT notebooks.



12.1 Validate Dataset Availability

This step confirms that all four engineered datasets exist as non-empty DataFrames before notebook-level validation begins.



Engineered dataset availability validated successfully.

	Dataset	Rows	Columns	Status
0	Sales Daily Dataset	573900	6	Passed
1	Calendar Dataset	1969	22	Passed
2	SKU Master Dataset	30490	16	Passed
3	Inventory Snapshot Dataset	573900	17	Passed



Validation Result

All four engineered datasets are available as non-empty DataFrames.

The notebook is ready for structural validation.



12.2 Validate Dataset Structure

This step verifies the exact schema, expected dimensions, and completeness of required fields across all engineered datasets.



Engineered dataset structures validated successfully.

	Dataset	Expected Rows	Actual Rows	Expected Columns	Actual Columns	Unexpected Missing	Status
0	Sales Daily Dataset	573900	573900	6	6	0	Passed
1	Calendar Dataset	1969	1969	22	22	0	Passed
2	SKU Master Dataset	30490	30490	16	16	0	Passed
3	Inventory Snapshot Dataset	573900	573900	17	17	0	Passed



Validation Result

All datasets matched their required dimensions and exact schemas.

No unexpected missing values were detected in required fields.



12.3 Validate Cross-Dataset Consistency

This step verifies Calendar coverage, SKU Master coverage, shared SKU-date keys, and demand consistency between Sales Daily and Inventory Snapshot.

Successful validation confirms that the engineered datasets form a consistent analytical data model.

Cross-Dataset Consistency Diagnostics

Sales-only SKU-date keys : 0
Inventory-only SKU-date keys : 0
Demand value mismatches : 0

Cross-Dataset Validation Summary

	Validation Check	Status
0	Calendar date key is unique	Passed
1	SKU Master key is unique	Passed
2	Sales SKU-date key is unique	Passed
3	Inventory SKU-date key is unique	Passed
4	Sales dates covered by Calendar	Passed
5	Inventory dates covered by Calendar	Passed
6	Sales and Inventory dates align	Passed
7	Sales and Inventory SKUs align	Passed
8	Sales SKUs covered by SKU Master	Passed
9	Inventory SKUs covered by SKU Master	Passed
10	Sales and Inventory keys align	Passed
11	Sales demand preserved in Inventory	Passed

✔ Cross-dataset consistency validated successfully.

✔ Validation Result

Calendar coverage, SKU Master coverage, shared SKU-date keys, and historical demand values were fully aligned.

The engineered datasets form a consistent analytical data model.

✔ 12.4 Validation Summary

This step summarizes the completed notebook-level validation stages and confirms readiness for downstream use.

Final Validation Summary

	Validation Stage	Status
0	Dataset Availability	Passed
1	Dataset Structure	Passed
2	Cross-Dataset Consistency	Passed

✔ All notebook-level validation checks passed.
Engineered datasets are ready for downstream Project FORESIGHT workflows.

✔ Validation Result

Dataset availability, structural integrity, and cross-dataset consistency were validated successfully.

Notebook 01 has produced a complete and internally consistent analytical data foundation.



13. Final Dataset Deliverables

The validated datasets were exported during their respective engineering workflows.

This section will register the final CSV deliverables, verify their availability and structure, and summarize the outputs produced by Notebook 01.



13.1 Register Final Dataset Deliverables

This step defines the final dataset names, validated in-memory sources, export locations, and date fields required for verification.

✓ Final dataset deliverables registered successfully.
Deliverables registered : 4
Output directory : C:\Users\hp\Project FORESIGHT\data\processed



Preparation Summary

Four final dataset deliverables were registered successfully with their validated in-memory sources and expected export locations.

The files are ready for centralized verification.



13.2 Verify Exported Files

This step verifies that every final CSV exists, contains data, and preserves the expected dimensions and schema.

✓ Exported dataset files verified successfully.

	Dataset	File Name	Rows	Columns	File Size (MB)	Status
0	Sales Daily Dataset	sales_daily_final.csv	573900	6	24.41	Passed
1	Calendar Dataset	calendar_final.csv	1969	22	0.15	Passed
2	SKU Master Dataset	sku_master_final.csv	30490	16	3.66	Passed
3	Inventory Snapshot Dataset	inventory_snapshots_final.csv	573900	17	42.30	Passed



Validation Result

All four exported CSV files were found and verified successfully.

Each file preserved its expected row count, schema, and missing-value structure and is available for downstream notebooks.



14. Notebook Conclusion

Notebook 01 completed the data-engineering and validation foundation for Project FORESIGHT.

Raw sales, calendar, pricing, and product data were transformed into four standardized analytical datasets. The final outputs were validated, exported, and verified for downstream use.

14.1 Notebook Achievements

Notebook 01 successfully:

- standardized identifiers, dates, schemas, and business measures;
- engineered daily sales, Calendar, and SKU Master datasets;
- created a documented inventory simulation using historical demand;
- validated structural and cross-dataset integrity;
- exported and verified four final CSV deliverables.

The complete data foundation is ready for analytical use.

14.2 Datasets Generated

Dataset	Purpose
Sales Daily Dataset	Daily SKU-level demand, revenue, price, and promotion records.
Calendar Dataset	Date attributes, events, seasons, weekends, and SNAP indicators.
SKU Master Dataset	Canonical SKU-store reference containing hierarchy, lifecycle, pricing, estimated cost, margin, and status attributes.
Inventory Snapshot Dataset	Simulated daily stock movements, replenishment activity, inventory status, estimated inventory value, and days of supply.

Together, these datasets form the validated analytical model for Project FORESIGHT.

14.3 Readiness for Notebook 02

All required datasets are available, structurally valid, internally consistent, and successfully exported.

Project FORESIGHT is ready to proceed to Notebook 02 for exploratory data analysis, visualization, and business-insight development.

Key Takeaway

Notebook 01 transformed the raw project data into a validated and reusable analytical foundation.

The final datasets are ready to support exploratory analysis, demand forecasting, simulated inventory analysis, and business decision-making in the remaining Project FORESIGHT notebooks.